

Compressing Global Dynamics of Deep Nonlinear Feature Learning

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Abstract

How much evolving information is needed to reproduce the training of a deep network? Networks that agree on every input can diverge once training resumes, so compressing a fitted function does not by itself compress its learning. We study this question while retaining depth, effective nonlinearity and feature learning. We organize descriptions of training by the state they retain and the accuracy it supports, distinguishing a simpler training regime from a compressed description of the same dynamics. Our main result compresses the learned response history: each neuron’s forward and backward histories are summarized by P shifted-Legendre moments in a clock that follows learning. For every fixed width and finite depth, the resulting autonomous system tracks the entire parameter trajectory on each prescribed finite horizon with error $O_T(P^{-1})$, or $O_T(P^{-2})$ with a joint response clock. Pairing the two memories makes the dynamical defect a product of projection errors, allowing control of their feedback through the network. The evolving internal state has $2(H - 1)mnP$ history coordinates for H hidden layers and m training samples, with an additional P^2 Gram state for the joint clock; the initialized matrices are retained. One fixed order serves the whole chosen interval. Thus accurate low-rank learned increments are generated online while features continue to move. Circle and MNIST experiments show that small orders closely reproduce the dense fitted predictor, including on unseen inputs. A common state–accuracy comparison places this construction alongside finite tangent hierarchies and numerical limiting-process descriptions.

1 Introduction

A missing model of learning. The Turing machine is the canonical model of computation. It is idealized—it has no limit on memory or precision—and it is simple enough to reason about, while physical computers, from the von Neumann architecture onward, are understood as its finite-resource implementations. One may argue that the Turing machine is a model of computing rather than of learning, and that learning still lacks its canonical idealized model. Deep neural networks trained by gradient descent are the obvious candidates for implementations of such a model, but the object they would implement is not yet known. The obstacle is that their training dynamics are enormously complex and have no simple representation.

Why look in the limit. A natural place to look for the idealized object is the limit of infinite width. Under feature-learning scalings of the type used in tensor-program and mean-field theory (Yang and Hu, 2021; Nguyen and Pham, 2023), width and learning rate are absorbed into the parametrization and the quantities we care about—predictions, kernels, the laws of hidden neurons—evolve deterministically. The limit is the network with no resource constraint, as the Turing machine is the computer with none. A finite network would then be understood as this limit plus a quantified deviation from it, a CLT-style picture in which each finite network combines its idealized limit with fluctuations around it.

Why compression. Going to infinite width to simplify seems paradoxical: the dynamical system grows with width and becomes infinite in the limit. Statistical physics resolves the paradox in other settings. A gas of 10^{23} molecules is simpler to describe than a gas of ten, because its macroscopic behaviour is captured by a few collective variables; mean-field theory replaces intricate particle interactions by smooth population fields. In neural training, eliminating explicit neuron indices can leave another large object: response histories, correlations between pairs of times, or evolving fields and operators. The question is then how much of that information is needed to reproduce learning to a prescribed accuracy. A finite network is Markovian in its weights; historical dependence appears when those weights are eliminated from a description. Compressing the resulting histories is one way to find a smaller sufficient state. We do not seek compression for its own sake. The variables that survive a controlled reduction are candidates for the variables that govern learning, and an autonomous compressed system offers a new way to study their interactions.

Training, not snapshots. Compressing a snapshot is easy: any representation of the current input–output map will do. But two networks that compute the same function on every input can have different Jacobians and backward responses, and once training resumes the discrepancy accumulates. An idealized model of learning must reproduce the whole trajectory.

What the model must keep. Several particularly tractable descriptions specialize to linear, shallow or frozen-feature regimes. Deep linear networks remove nonlinear activations (Saxe et al., 2014); one-hidden-layer mean-field models remove interactions between multiple learned hidden layers (Chizat and Bach, 2018; Mei et al., 2018; Sirignano and Spiliopoulos, 2019); and the neural tangent kernel retains nonlinear input features while freezing their leading-order evolution (Jacot et al., 2018; Lee et al., 2019; Chizat et al., 2019). These are simplifications of the training process being studied, not claims about an inability to represent a complicated final function. We keep all three: *depth* (more than one hidden layer), *nonlinearity* that stays effective during training and in the limit, and *feature learning* (hidden representations move at leading order). Together with the population limit, this contract preserves what is hardest about deep learning while making it a single well-defined object.

The gap: accuracy versus retained state. Tensor Programs remove explicit width through a limiting recursion that includes past responses (Yang and Hu, 2021); nonlinear dynamical mean-field theory (DMFT) describes their evolving covariances and responses through two-time kernels (Bordelon and Pehlevan, 2022). The Neural Tangent Hierarchy (NTH) instead organizes current outputs and higher tangent kernels, allowing finite truncations that evolve features (Huang and Yau, 2020). These descriptions economize on different information. A common question is therefore *how much evolving state suffices to approximate a specified training trajectory over a complete interval?*

We address this question by compressing the learned response history itself. The representation uses a fixed number of evolving coordinates per neuron and training sample, and reconstructs the network through which those coordinates continue to evolve. Our guarantee concerns the full parameter trajectory at fixed width; it does not require small feature movement. The broader comparison in Section 7 puts width, hierarchy order and temporal resolution on one state–accuracy axis, while making the assumptions needed to match their population targets explicit.

Results. Section 4 makes compression of dynamics precise and distinguishes three classes of state: scalars, populations and operators. We then show the following.

1. *Exact descriptions expose closure obstacles* (Section 5). An operator description of a deep linear benchmark, a hierarchy of current population observations, and a restricted scalar nonclosure result distinguish an exact representation from a finite collection of sufficient coordinates.
2. *Training history is compressible in time* (Section 6). For networks of any depth, replacing each neuron’s forward and backward history by P Legendre moments in a learning-speed clock gives an autonomous system whose state does not grow with training time and whose trajectory is within $O_T(P^{-1})$ of gradient flow, or $O_T(P^{-2})$ with a joint clock (Theorems 6.1 and 6.2). The difficulty is feedback: the memorized signals depend on the approximation itself.
3. *A structural consequence* (Corollary 6.3). Every learned weight increment is, on a finite horizon, within $O(1/P)$ of a rank- mP matrix built from the network’s own response moments.
4. *Evolving memories are what matter* (Section 8). Small orders reproduce dense fitted predictions around the input circle and on held-out MNIST inputs. In the tested circle comparisons they outperform frozen dictionaries and directly trained low-rank corrections of matched size.
5. *A common resource question* (Section 7). We distinguish methods that simplify the training regime from representations of deep feature learning, and compare the latter by moving state, fixed resources and prediction accuracy. Conditional population scalings make the cost of width and history resolution explicit, without identifying them as a single approximation.

The theorems hold at every fixed width. The population limit is the motivation and gives an interpretation of the result (Section 6.6), and a precise conjecture (Conjecture 6.4) states what a population version would be; it is not proved here. Appendix C records an earlier, weaker closure of the population flow itself, in which the learned middle layer is confined to a frozen span of initialization observables; it converges without a rate, and its limitations motivated the evolving memories of the main result.

2 Related work

We distinguish the training process described by a theory from the information its representation retains. Section 7 develops the corresponding quantitative comparison.

Regime-specific simplifications. Exact solutions of deep linear networks keep depth and feature learning but not nonlinearity (Saxe et al., 2014). Wasserstein and mean-field limits describe one-hidden-layer networks as the gradient flow of a measure over neurons (Chizat and Bach, 2018; Mei et al., 2018; Sirignano and Spiliopoulos, 2019); they keep nonlinear feature learning but not depth. The neural tangent kernel and lazy training keep depth and nonlinearity but freeze the features, reducing training to a fixed kernel (Jacot et al., 2018; Lee et al., 2019; Chizat et al., 2019). Each isolates an important mechanism, but its accuracy theorem alone does not approximate the deep feature-learning flow studied here.

Descriptions of deep feature learning. Tensor Programs IV proves feature-learning limits at each fixed number of steps, with a recursion that includes variables indexed by every earlier step (Yang and Hu, 2021). Feature-learning DMFT evolves two-time activation, gradient and response kernels; its full-grid implementation has quadratic kernel storage in sample count times time-grid length (Bordelon and Pehlevan, 2022). The numerical work also depends on sampling, response

estimation and self-consistency iterations. Its finite-width extension derives perturbative fluctuations (Bordelon and Pehlevan, 2023); the general nonlinear field construction and its numerical solver have different proof obligations from a rigorous finite-width convergence theorem. Multilayer mean-field theory and Neural Feature Flow evolve fields on neuronal ensembles (Nguyen and Pham, 2023; Fang et al., 2020). These keep all three properties and provide descriptions of deep feature learning under their respective parameterizations and assumptions. Their state can be independent of explicit width while retaining temporal information or infinite-dimensional fields. Response memory keeps a finite-width population and compresses its learned history; its limiting population interpretation is stated separately.

Finite hierarchies. The Neural Tangent Hierarchy derives an exact hierarchy of outputs and higher tangent kernels and truncates it at order p (Huang and Yau, 2020). Higher truncations evolve the tangent kernel and capture feature learning. In the direct implementation, predictions and kernels below order p contribute $\sum_{j=1}^{p-1} m^j$ moving scalars, while the m^p entries of the top tensor are fixed. Its published approximation theorem controls a width-dependent regime; transfer to canonical feature learning requires bounds on the matched higher kernels. Appendix E.2 gives that hierarchy and a conditional finite-horizon comparison. Response memory provides an alternative refinement coordinate: history moments with a polynomial order guarantee for the full fixed-width parameter trajectory.

Memory and low rank. HiPPO provides online Legendre projection of a given signal (Gu et al., 2020); we apply it to endogenous, coupled training histories and prove stability of the resulting feedback loop. Low-rank training methods constrain the learned increment to evolving factors (Zhao et al., 2023); response-memory factors are causal summaries of the actual forward and backward histories, and a directly trained factorization of matched rank performs far worse (Section 8).

3 Setting

Fix m training pairs (x_a, y_a) with $x_a \in \mathbb{R}^d$, width n and $H \geq 2$ hidden layers:

$$\begin{aligned} z_a^{(1)} &= W^{(1)} x_a / \sqrt{d}, & h_a^{(1)} &= \phi(z_a^{(1)}), \\ z_a^{(\ell)} &= W^{(\ell)} h_a^{(\ell-1)}, & h_a^{(\ell)} &= \phi(z_a^{(\ell)}), \quad 2 \leq \ell \leq H, \\ f_a &= \frac{w^\top h_a^{(H)}}{n}, & r_a &= f_a - y_a, \quad \rho = \left(\frac{1}{m} \sum_a r_a^2 \right)^{1/2}. \end{aligned} \tag{1}$$

The activation ϕ may differ between layers; the loss is $\mathcal{L} = \rho^2$. The backward responses, which exclude the residual, are $\delta_a^{(H)} = w \odot \phi'(z_a^{(H)})$ and $\delta_a^{(\ell)} = \phi'(z_a^{(\ell)}) \odot (W^{(\ell+1)})^\top \delta_a^{(\ell+1)}$. With block mobilities $(n, 1, \dots, 1, n)$, gradient flow reads

$$\dot{W}^{(1)} = -\frac{2}{m} \sum_a r_a \delta_a^{(1)} \frac{x_a^\top}{\sqrt{d}}, \quad \dot{W}^{(\ell)} = -\frac{2}{nm} \sum_a r_a \delta_a^{(\ell)} (h_a^{(\ell-1)})^\top \quad (2 \leq \ell \leq H), \quad \dot{w} = -\frac{2}{m} \sum_a r_a h_a^{(H)}. \tag{2}$$

We write $\theta = (W^{(1)}, \dots, W^{(H)}, w)$ for all parameters and $\dot{\theta} = F(\theta)$ for (2). Hidden entries are initialized as $\mathcal{N}(0, 1/n)$, first-layer entries as $\mathcal{N}(0, 1)$, and the stored readout is small.

Population flow. As $n \rightarrow \infty$, the neurons of layer ℓ become a probability space Ω_ℓ with expectation $\mathbb{E}_\ell$: hidden coordinates $z^{(\ell)}, h^{(\ell)}$ become random variables in $L^2(\Omega_\ell)$, the first-layer weights a random row $W^{(1)} \in L^2(\Omega_1; \mathbb{R}^d)$, the readout $w \in L^2(\Omega_H)$, and each hidden matrix a bounded operator $W^{(\ell)} : L^2(\Omega_{\ell-1}) \rightarrow L^2(\Omega_\ell)$ used together with its true adjoint. The same equations (2) hold with $\frac{1}{n} \sum_i$ replaced by $\mathbb{E}_\ell$ and outer products $\delta h^\top / n$ by the rank-one operators $g \mapsto \delta \mathbb{E}[h g]$. Such limits are established in several scopes—for fixed depth on a short horizon, and globally in special models—with finite-width gradient flow converging to them on compact horizons; a global theorem for general data and depth is open. Where we need the population flow we state its existence as part of the model.

4 Compressing a dynamical system

Exact and approximate; local and global. An exact compression reproduces the reference trajectory; an approximate one reproduces it up to a controllable error. A compression may also match the current state and velocity while its small errors compound until the trajectories separate; we ask for the stronger, global notion.

Definition 4.1 (Global approximation on compact horizons). Autonomous systems indexed by an order N , with trajectories θ_N , *globally approximate* a reference flow θ if for every $T < \infty$

$$\sup_{0 \leq t \leq T} \|\theta_N(t) - \theta(t)\| \rightarrow 0 \quad (N \rightarrow \infty).$$

The rate in N quantifies the trade-off between the size of the compressed state and its accuracy.

Three classes of state.

1. A *scalar ODE* evolves finitely many real numbers, independently of width and training time.
2. A *population* (or field) ODE evolves finitely many functions, random variables or probability laws on fixed spaces—for example the law of a neuron’s weights. Its particle implementation stores one value per particle and field; the number of objects is finite, their degrees of freedom are not.
3. An *operator ODE* evolves operators between population spaces, the limits of matrices. A matrix records how *each* row relates to *each* column, and this row–column coupling cannot in general be captured by finitely many one-neuron marginal statistics alone.

A finite number of named fields or operators need not contain finitely many scalars, so every compression claim below names both the type and the size of the retained state. We also distinguish *moving state*, updated during training or a numerical solve, from *fixed resources*, such as initialized matrices or frozen kernel coefficients. The same convention applies to every method; total storage and arithmetic count both. For a prescribed horizon and accuracy, we ask how much moving state a specified representation needs, rather than interpreting a finite list of mathematical objects as a numerical size.

5 Exact descriptions and obstacles to finite closure

Before allowing errors we ask what exact capture of the population flow can look like. The following examples show why writing an exact evolution and finding a small autonomous state are distinct problems.

An exact operator description. A natural description of the population flow is an operator ODE: (2) read with the hidden matrices $W^{(\ell)}$ as operators. In a deep linear benchmark—one input, three linear hidden layers, order-one initialization—this can be proved completely: the learned part of the whole network becomes a single trace-class operator that evolves by an explicit autonomous cubic equation driven by the residual, and finite gradient flow and gradient descent converge to it on every compact horizon (Appendix D). The description is exact and consists of one operator. This identifies an autonomous limiting state, while its numerical complexity depends on how that operator is represented. Dense coordinates and stored response histories are two possible implementations; operator structure may allow others.

Current observations generate a hierarchy. For a two-hidden-layer population flow on the circle (Appendix C), one can describe the dynamics exactly through *current* observations: joint laws, on each neuron population, of finitely many expressions built from the current weights, the initialization, ϕ , bounded products and applications of $W^{(2)}$ and its adjoint. The derivative of each finite level of this hierarchy is expressed through a finite higher level, and the complete hierarchy determines predictions on the whole input space and a unique continuation from every reachable state, without storing the past. Differentiating a finite level introduces higher observations, so the hierarchy alone does not provide a finite autonomous closure.

A restricted scalar obstruction. In the same linear benchmark, no encoder formed from a fixed finite list of tensor contractions of the weights with width-independent bounded degree admits a polynomial autonomous scalar ODE and polynomial readout that reproduce the output dynamics on an open set of states at all large widths (Appendix D). The reason is that repeated differentiation of the output forces ever larger connected contractions: the $(2j - 1)$ st derivative contains

$$\frac{1}{n} (W^{(1)})^\top \left((W^{(2)})^\top (W^{(3)})^\top W^{(3)} W^{(2)} \right)^j W^{(1)},$$

of degree $4j + 2$, which no bounded-degree alphabet can express. The result is restricted to this encoder and readout class. It shows that even a linear network can generate an unbounded hierarchy of interactions under training; it does not exclude arbitrary encoders or trajectory-specific reductions.

Between the two. Whether the general exact operator can be replaced by finitely many population fields remains open. Instantaneous low-rank updates do not settle the question: their directions change, and an integral over time need not have a uniformly bounded rank. Approximation asks a more constructive question: can finitely many evolving fields generate the learned action to controlled accuracy even when those directions continue to move? The response memories below do so at each fixed width.

6 Approximate compression with rates: response memories

We now ask what can be gained by accepting a controlled error, for networks of any depth and with explicit order-versus-error rates.

6.1 The idea

Integrating (2), every hidden matrix is

$$W^{(\ell)}(t) = W_0^{(\ell)} - \frac{2}{nm} \sum_a \int_0^t r_a(s) \delta_a^{(\ell)}(s) (h_a^{(\ell-1)}(s))^\top ds. \quad (3)$$

The learned part is an integral of backward against forward responses. Storing it means storing the dense matrix or the whole response history. But to predict where a moving ball goes next we need its position, velocity and acceleration, not its past positions. If each response history is regular in time, a few coefficients should summarize it. Two things must hold: the summary must be *closed*, so that it evolves autonomously, and *accurate* over the whole of training, which alternates between bursts of learning and plateaus.

6.2 Closure: shifted Legendre moments

Let p_k be the shifted Legendre polynomials on $[0, 1]$, normalized by $p_k(1) = 1$ and $\int_0^1 p_j p_k = \mathbf{1}_{j=k}/(2k+1)$. Their derivatives reduce to lower modes,

$$x p'_k(x) = k p_k(x) + \sum_{j < k} (2j+1) p_j(x), \quad (4)$$

and this is what makes the moment system close exactly. The identity underlies online polynomial memory in HiPPO (Gu et al., 2020); the new element is that the memorized signals are the network's own forward and backward responses, coupled back through the reconstructed network.

6.3 Accuracy: a clock that follows learning

A polynomial basis is accurate only if history is spread evenly over its window, and in physical time it is not. We therefore measure history in a clock τ that runs at the speed of learning,

$$\dot{\tau} = \rho, \quad \tau(0) = 1 : \quad (5)$$

fast while the network learns, slow as learning stalls. The unit prefix $[0, 1]$ carries the initial forward responses and no backward response. For each hidden link $\ell \geq 2$ and sample a we store the first P Legendre moments, in the clock coordinate, of the forward response $h_a^{(\ell-1)}$ and of the residual-weighted backward response $r_a \delta_a^{(\ell)}/\rho$:

$$\bar{h}_{a,k}^{(\ell-1)} = \int_0^\tau h_a^{(\ell-1)} p_k(\xi/\tau) d\xi, \quad \bar{\delta}_{a,k}^{(\ell)} = \int_0^\tau \frac{r_a \delta_a^{(\ell)}}{\rho} p_k(\xi/\tau) d\xi, \quad 0 \leq k < P,$$

where ξ is the clock value at which each response was recorded. By (4) these moments obey the closed system

$$\begin{aligned} \dot{\bar{h}}_{a,k}^{(\ell-1)} &= \rho h_a^{(\ell-1)} - \frac{\rho}{\tau} \left[k \bar{h}_{a,k}^{(\ell-1)} + \sum_{j < k} (2j+1) \bar{h}_{a,j}^{(\ell-1)} \right], \\ \dot{\bar{\delta}}_{a,k}^{(\ell)} &= r_a \delta_a^{(\ell)} - \frac{\rho}{\tau} \left[k \bar{\delta}_{a,k}^{(\ell)} + \sum_{j < k} (2j+1) \bar{\delta}_{a,j}^{(\ell)} \right], \end{aligned} \quad (6)$$

with $\bar{h}_{a,0}^{(\ell-1)}(0) = h_a^{(\ell-1)}(0)$ and all other moments zero, and the hidden matrices are reconstructed as

$$\widehat{W}^{(\ell)} = W_0^{(\ell)} - \frac{2}{nm\tau} \sum_{a=1}^m \sum_{k=0}^{P-1} (2k+1) \bar{\delta}_{a,k}^{(\ell)} (\bar{h}_{a,k}^{(\ell-1)})^\top. \quad (7)$$

All responses in (6) are evaluated in the current *reconstructed* network, and the first layer and readout follow (2). The system is autonomous and never divides by ρ . Equivalently, with Π_P the

$L^2(0, \tau)$ projection onto polynomials of degree below P ,

$$\widehat{W}^{(\ell)} = W_0^{(\ell)} - \frac{2}{nm} \sum_a \int_0^\tau \left(\Pi_P \frac{r_a \delta_a^{(\ell)}}{\rho} \right) (\Pi_P h_a^{(\ell-1)})^\top d\xi, \quad (8)$$

and since cross terms vanish by orthogonality, the error in each matrix update is a *product* of a forward and a backward projection error.

The joint feature-learning clock. The learning-speed clock makes forward histories smooth but not every backward history. A second clock advances with the residual *plus* the speed of all forward and backward responses,

$$\dot{\tau} = \rho + \left\| \frac{d}{dt} (h_a^{(\ell-1)}, r_a \delta_a^{(\ell)} / \rho)_{\ell, a} \right\|_2, \quad \tau(0) = 1, \quad (9)$$

so that the whole response path moves at unit speed at most in the clock coordinate. History is placed by this clock but still weighted by the residual mass ρdt , so the Legendre modes are no longer orthogonal and one shared $P \times P$ Gram matrix is tracked alongside the moments; the reconstruction uses its inverse, the defect keeps the product form, and the clock speed can be evaluated explicitly by one directional forward and backward pass (Appendix B).

6.4 Global guarantees

Theorem 6.1 (Learning-speed clock). *Fix finite width, depth $H \geq 2$, arbitrary finite data and a finite initialization, and let every activation be in $C_{\text{loc}}^{1,1}(\mathbb{R})$. For every $T < \infty$ there are $C(T)$ and $P_0(T)$, independent of P , such that for $P \geq P_0(T)$ the closure exists uniquely on $[0, T]$ and*

$$\sup_{0 \leq t \leq T} \left\| \widehat{\theta}_P(t) - \theta(t) \right\| \leq \frac{C(T)}{\sqrt{P(P+1)}}.$$

The rate extends to hidden responses and to predictions on any bounded set of unseen inputs. If ϕ and ϕ' are bounded (tanh, sigmoid, arctangent, sine), it holds for every $P \geq 1$. No assumption of successful fitting, Gaussian initialization, input geometry or label structure is made.

Theorem 6.2 (Joint feature-learning clock). *Under the setting of Theorem 6.1, assume the activations of layers $\ell \geq 2$ are in $C_{\text{loc}}^{2,1}$ and $\rho(0) > 0$. For every T there are $P_0(T)$ and $C(T)$ such that for $P \geq P_0(T)$ the closure exists uniquely on $[0, T]$, its clock stays bounded independently of P , and $\sup_{t \leq T} \left\| \widehat{\theta}_P(t) - \theta(t) \right\| \leq C(T)/[P(P+1)]$.*

What separates this from approximating a given signal, as in HiPPO, is feedback: the histories being memorized are produced by the network reconstructed from the memories, so approximation errors change the signals they approximate. Both proofs (Appendices A and B) rest on three facts: an exact identity for the growth of the projection error of each history; the product structure of the defect; and the Legendre tail bound $\|v - \Pi_P v\|^2 \leq \tau^2 \|v'\|^2 / (4P(P+1))$. With the learning-speed clock the forward histories have P -independent derivative bounds while backward histories only need bounded mass, giving $O(P^{-1})$; with the joint clock the whole path is unit speed once the clock is shown to be bounded, both factors contribute, and the rate is $O(P^{-2})$. Two steps carry the difficulty. In the learning-speed clock, bounding forward histories through the layers requires the reconstructed velocity, whose memory endpoint term grows like P ; it is cancelled exactly by the P^{-1} projection factor, which is why the rate does not degrade with depth. In the joint clock,

the clock itself must be bounded without assuming it, which we do through an order-independent bound on the total variation of the trajectory. Gronwall’s inequality and a first-exit argument then transfer the accumulated defect to the trajectory.

Corollary 6.3 (Learned increments are low-rank in their own history). *Under Theorem 6.1, for every T and every $P \geq P_0(T)$, each dense learned increment $W^{(\ell)}(t) - W_0^{(\ell)}$, $0 \leq t \leq T$, is within $C(T)/\sqrt{P(P+1)}$ in Frobenius norm of the matrix $-\frac{2}{nm\tau} \sum_{a,k < P} (2k+1) \bar{\delta}_{a,k}^{(\ell)} (\bar{h}_{a,k}^{(\ell-1)})^\top$, which has rank at most mP and is built from the moments of the closure’s own responses. Under Theorem 6.2, the joint-clock reconstruction, of rank at most $m(P+1)$, is within $C(T)/[P(P+1)]$.*

The corollary follows from the theorems because the reconstruction (7) has rank at most mP . Its content is not that learned increments admit low-rank approximations—a truncated singular value decomposition would supply those, but only from the dense matrix itself. It is that a good approximation is *generated autonomously and online* from the network’s own forward and backward history, with a rank that does not grow with time: on a given horizon, one fixed order serves every time inside it, however much the features have moved. The statement is informative only while $mP < n$, and its constant depends on width.

6.5 Cost

With H layers, width n , m samples and order P , each learned $n \times n$ increment is carried by $2mP$ vectors of length n . The moving internal state is $2(H-1)mnP$ numbers against $(H-1)n^2$ for the dense learned matrices; it does not grow with training time and is smaller whenever $2mP < n$. The initialized matrices, $(H-1)n^2$ numbers, are kept as a fixed source. Per step, applying $W_0^{(\ell)}$ and its transpose costs $O(mn^2)$ per layer as in dense training, the learned correction $O(m^2nP)$ per layer, and the moment update $O(HmnP)$; the joint clock adds an $O(P^3)$ Gram factorization and $O(HmnP^2)$ solves. At fixed width and horizon, suppressing their constants, trajectory error ε requires $O(Hmn\varepsilon^{-1})$ history numbers with the learning-speed clock and $O(Hmn\varepsilon^{-1/2} + \varepsilon^{-1})$ with the joint clock.

What is compressed is time, not samples or neurons. The state grows linearly with the number of training samples, and once $2mP \geq n$ it exceeds the dense learned matrices; learning from a continuum of inputs would need a further representation of the input dependence. Nor is this a runtime improvement: because $W_0^{(\ell)}$ still acts densely, the closure ran somewhat slower than dense training in our experiments.

6.6 From finite networks to the population

The theorems hold at every fixed width, for every realized initialization. At the population level they suggest the following picture, which we state as an interpretation, not a result. For each layer and sample, the P forward and P backward moments are random variables over neurons, and the learned action of $W^{(\ell)}$ becomes $g \mapsto -\frac{2}{m\tau} \sum_{a,k} (2k+1) \bar{\delta}_{a,k}^{(\ell)} \mathbb{E}_{\ell-1}[\bar{h}_{a,k}^{(\ell-1)} g]$: finitely many neuron-level memory fields, coupled through the initialized operator. A width- n network would then be a Monte Carlo simulation of this system. What is proved is only a diagonal statement: if finite-width flows converge to a population flow on $[0, T]$, choosing $n_j \rightarrow \infty$ and then P_j large enough at each width yields compressed systems converging to it. The natural next theorem is the following.

Conjecture 6.4 (Fixed-order population closure). *For each fixed P , as $n \rightarrow \infty$ the order- P response-memory closure converges on compact horizons to a limiting system consisting of the initialized operators and finitely many neuron-level memory fields per layer and sample; and this*

limit tracks population gradient flow with error $O_T(P^{-1})$, or $O_T(P^{-2})$ with the joint clock, uniformly in width.

The limit in the conjecture would be a time-local, finite-order description of deep nonlinear feature learning in the population limit: the kind of object the introduction asks for.

7 Accuracy and state size across descriptions of training

A common resource question connects otherwise different theories: how much evolving information suffices to approximate the same learning process? The first distinction is between simplifying that process and compressing its description. Linear, shallow and frozen-feature models give tractable accounts of important regimes. Dense training, tangent hierarchies, limiting-response solvers and response memory instead provide different representations to compare when depth, nonlinearity and feature learning are retained (Table 1).

A shared target. Fix the architecture, data, initialization law (including the readout), canonical scaling and a population flow f_∞ on $[0, T]$. For a test probability measure Q , write

$$E_Q(\hat{f}; T) = \sup_{t \leq T} \left\| \hat{f}(t, \cdot) - f_\infty(t, \cdot) \right\|_{L^2(Q)}, \quad \mathcal{E}_Q = (\mathbb{E} E_Q^2)^{1/2}. \quad (10)$$

The expectation covers initialization or numerical sampling. Uniform angular Q gives whole-circle prediction RMSE. The reverse triangle inequality bounds the difference between the two predictors' test RMSEs by E_Q . This common observable allows comparison with methods that predict the learned function without reconstructing all trained weights; our parameter theorem also supplies the latter.

Matched-regime idealizations. The table separates representation counts from conditional population accuracy. For the quantitative comparison, assume the specified population target exists, the dense error is bounded by $A_T/\sqrt{n}$, and the closure estimates are uniform in width in a normalized prediction-controlling norm. Then coupling the closure and dense network through the same initialization gives

$$\mathcal{E}_{\text{dense}} \leq A_T n^{-1/2}, \quad \mathcal{E}_q \leq A_T n^{-1/2} + B_{q,T} P^{-q}, \quad q = 1, 2. \quad (11)$$

Here $q = 1$ denotes the learning-speed clock and $q = 2$ the joint clock. The root-width scale is supported by perturbative fluctuation theory (Bordelon and Pehlevan, 2023); a general matched rate and width-uniform closure constants are additional hypotheses, not conclusions of Theorems 6.1 and 6.2. All constants can depend on data, m, H, T and Q . Numerical limiting-process bounds additionally require stable sampling, time discretization and self-consistency. These assumptions are explicit because they determine what can be compared, rather than supplying a universal ordering of methods.

What each refinement buys. NTH evolves features through successively higher tangent kernels. Its same-target truncation error $\eta_p(T)$ depends on their growth; the published width-dependent theorem cannot simply be transferred to another scaling. Appendix E.2 derives a feedback bound for the canonical hierarchy and explains why it is not a time-Taylor truncation. For TP/DMFT we specify a sampled full-grid response solver (Yang and Hu, 2021; Bordelon and Pehlevan, 2022): K is the number of time steps, S the number of paths, r the numerical time order and η the solver error. Streaming paths leaves $O(Hm^2 K^2)$ kernel state while increasing sampling work. Thus S is not a finite-network width, and a numerical time grid is not the physical horizon T .

Table 1: Descriptions of training on a common state–accuracy axis. The upper block changes a property of the target dynamics; its rows are not general accuracy refinements of the specified deep feature-learning flow. The lower block gives direct representation counts and conditional population-prediction laws explained in the text. Moving state excludes fixed resources consistently. Fixed input dimension and lower-order outer weights are suppressed; $m \geq 2$ for the direct NTH count.

Description	Moving state	Fixed resources	Error mechanism or regime change
<i>Simplifying the training regime</i>			
Deep linear	—	—	Removes nonlinear activations
One-hidden-layer mean field	—	—	Removes multilayer hidden interactions
Frozen NTK	m	m^2 kernel; test evaluator	Feature-freezing bias relative to rich training
<i>Representations of deep nonlinear feature learning</i>			
Dense width n	$O(Hn^2)$	—	$A_T n^{-1/2}$
NTH through p	$\sum_{j=1}^{p-1} m^j$	m^p top tensor; mixed-kernel functions	$\eta_p(T)$ from higher-kernel bounds
Sampled TP/DMFT, full grid	$O(Hm^2 K^2)$	Model and input covariances	$A_{D,T} S^{-1/2} + B_{D,T}(T/K)^r + \eta$
Learning-speed memory	$O(HmnP)$	$O(Hn^2)$ initialization	$A_T n^{-1/2} + B_{1,T}/P$
Joint response memory	$O(HmnP + P^2)$	$O(Hn^2)$ initialization; prefixes	$A_T n^{-1/2} + B_{2,T}/P^2$

Size at prescribed accuracy. Under (11), optimizing width and order at fixed task gives the following leading moving-memory scalings:

$$\begin{aligned}
M_{\text{dense}}^*(\varepsilon) &\asymp H A_T^4 \varepsilon^{-4}, \\
M_{\text{old}}^*(\varepsilon) &\asymp H m A_T^2 B_{1,T} \varepsilon^{-3}, \quad M_{\text{joint}}^*(\varepsilon) \asymp H m A_T^2 \sqrt{B_{2,T}} \varepsilon^{-5/2}.
\end{aligned} \tag{12}$$

These optimize sufficient error bounds for the stated representations; Appendix E.1 gives the allocations and the corresponding conditional NTH and numerical TP/DMFT costs. They are not lower bounds on all algorithms within a framework. Keeping dense initialization leaves a leading ε^{-4} contribution to our total storage and arithmetic. The established gain is in the evolving learned state; improving total cost also requires treating the fixed environment.

This comparison separates width resolution from history resolution. At fixed width the response-memory rates are proved, and the state is linear in the number of retained moments. The population calculation predicts how that gain combines with width approximation. It does not assume equal or small horizon constants across methods: one fixed order serves every time inside a chosen interval, but increasing that interval may require more moments. Appendix E records the derivations, per-step costs and test-function evaluation requirements so that these common-axis comparisons can be refined without changing the underlying question.

Table 2: Whole-circle RMS difference between response memory and dense gradient flow. Left: two hidden tanh layers, width 2048 (Figure 2). Right: three hidden tanh layers, width 4096 (Figure 1).

Task	two layers			three layers		
	$P = 1$	$P = 3$	$P = 7$	$P = 1$	$P = 2$	$P = 3$
Two outliers, alternating	0.236	0.0730	0.00466	0.108	0.0238	0.0315
Quadrant, alternating	0.341	0.0456	0.00155	0.275	0.0284	0.0585
Quadrant, paired	0.0193	0.00282	0.00012	0.0122	0.00371	0.00121
Quadrant, center/edges	0.0456	0.00488	0.00131	0.0790	0.0257	0.00725
Equally spaced, mixed	0.0124	0.00053	4×10^{-6}	0.0150	0.00241	0.00028

8 Experiments

We integrate finite networks and compare response memory with dense gradient flow from the same initialization, using the learning-speed clock unless stated. With eight labelled points on a circle, many functions fit the labels, so agreement at training points says little; we compare the whole function that training selects, on 8192 angles, at each model’s first crossing of training MSE 0.001. The five tasks are chosen to be hard: alternating labels, clustered points and large unlabelled arcs. These are comparisons of fitted functions at matched loss, with one initialization per task, not same-time trajectory errors: they test the closure as an estimator of what dense training produces, not the rates of the theorems. Figures 1 and 2 show the fitted functions for both depths in radial coordinates.

Circle tasks. In two layers the error falls by 1.5–3 orders of magnitude from $P = 1$ to $P = 7$ on every task, faster than the P^{-1} trajectory bound (Figure 3); the theorem is an upper bound on a different quantity, and we do not read a rate off these five points. In three layers $P = 2$ and $P = 3$ have whole-circle RMS differences below 0.06, and below 0.01 on three tasks, while the hidden layers move substantially: relative activation displacement is 0.35–0.65, 0.38–0.74 and 0.46–0.68 in the three layers. Order is not monotone at fixed width: $P = 2$ beats $P = 3$ on both alternating tasks. The moving state at $P = 3$ is about 3 MiB against 256 MiB for the dense learned matrices.

Evolving memories versus fixed or trained bases. The corollary says that learned increments are low-rank *in their own history*. Other low-rank bases do not work. Frozen dictionaries built at initialization reach RMS 0.55–1.38 on the outlier task with about 45 vectors, against 0.073 for 48 response moments. A directly trained low-rank correction of the middle matrix, with factors of matched rank and the same $W_0^{(2)}$, lost in all 29 valid comparisons, 28 of them by more than a factor of two; at rank 56 on the outlier task the errors were 0.0047 against 0.52 and 0.25 for two factor seeds.

MNIST. Predictions of response memory and dense training on held-out images lie on the diagonal (Figure 4), with identical test accuracy (94.7%). With 1000 training images and width 1024 the differences are 0.019, 0.0019 and 0.0012 for $P = 1, 2, 3$, with 97.5% accuracy for dense, $P = 2$ and $P = 3$.

Two clocks. The joint clock’s better rate is asymptotic. At $P = 5$ it improved some cases (tanh alternating quadrant $0.0066 \rightarrow 0.0035$; GELU outliers $0.0020 \rightarrow 0.0010$) and was much worse on another (GELU alternating quadrant $0.013 \rightarrow 1.00$, reduced to 0.10 by halving the step). Over an

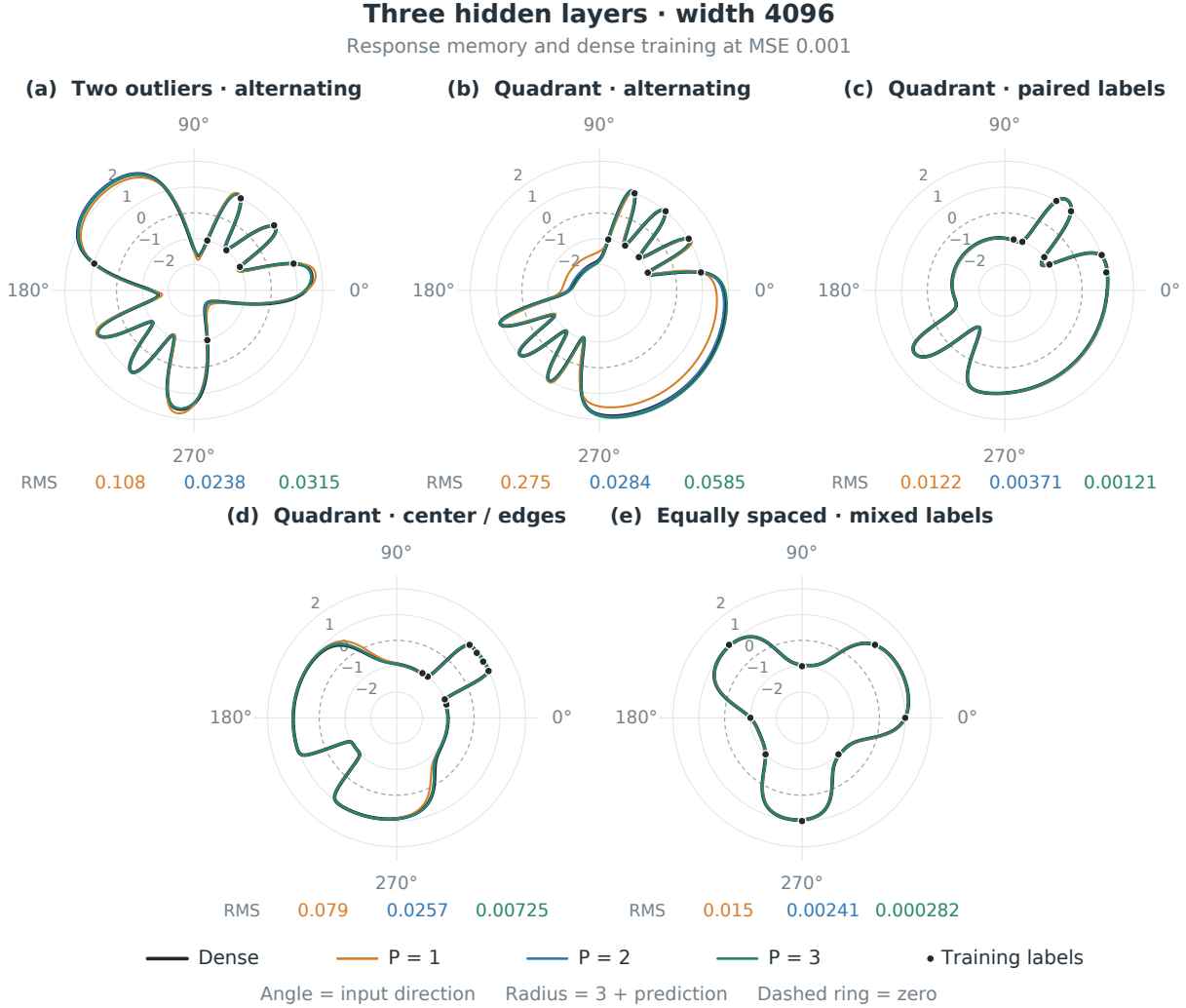


Figure 1: Fitted circle predictions for three hidden tanh layers, width 4096, and response-memory orders $P = 1, 2, 3$. Models are evaluated at their respective training-MSE 10^{-3} endpoints. Angle represents input direction; radius is $3 + f(\theta)$, with radial ticks labelled by $f(\theta)$ and a dashed zero-output ring. Black curves show dense gradient flow, coloured curves show response memory, and black dots show training labels. Coloured numbers give RMS prediction differences from the dense network on 8192 uniform circle points, in legend order.

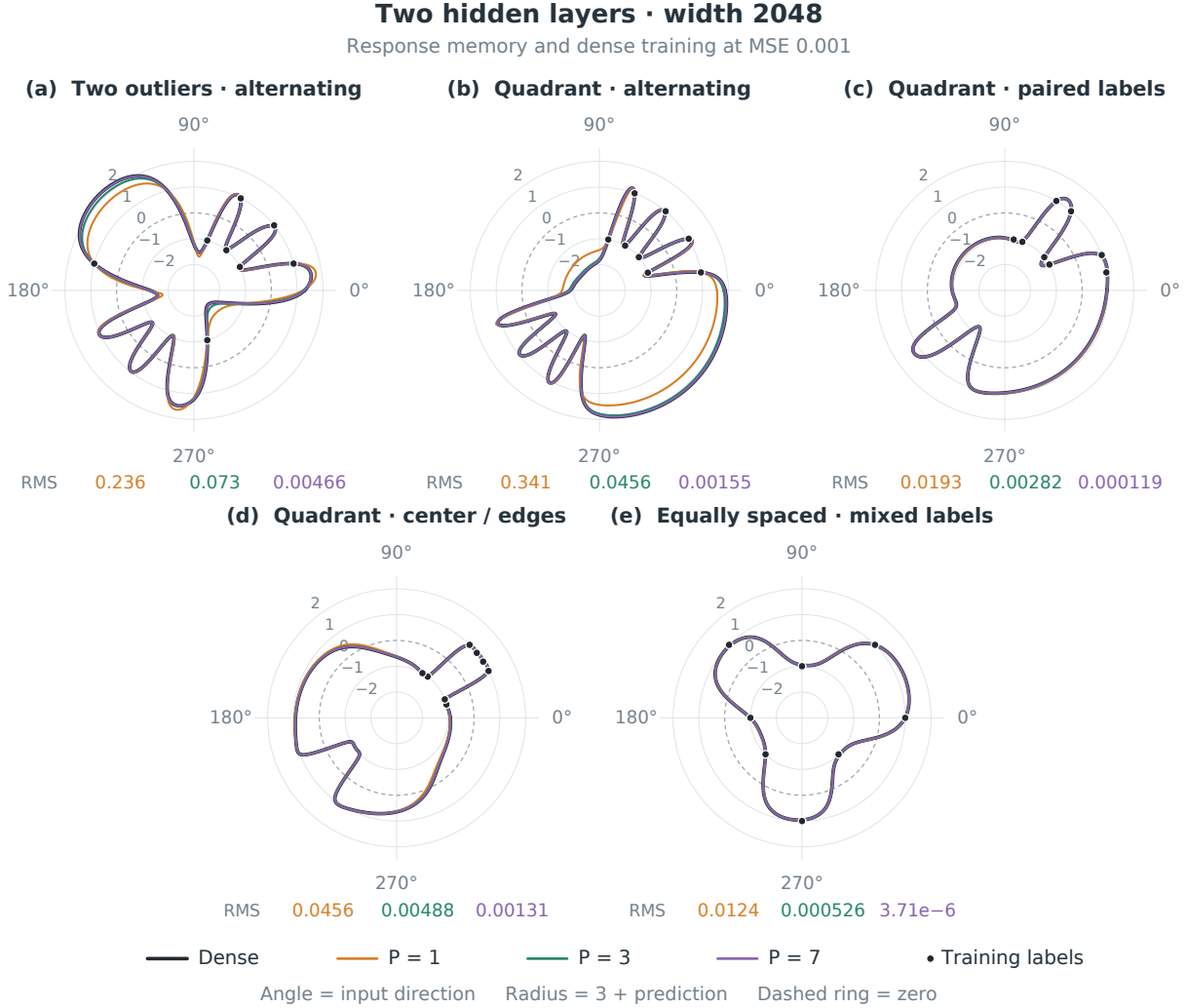


Figure 2: Fitted circle predictions for two hidden tanh layers, width 2048, and response-memory orders $P = 1, 3, 7$, using the dense references in Table 2. Models are evaluated at their respective training-MSE 10^{-3} endpoints. Angle represents input direction; radius is $3 + f(\theta)$, with radial ticks labelled by $f(\theta)$ and a dashed zero-output ring. Black curves show dense gradient flow, coloured curves show response memory, and black dots show training labels. Coloured numbers give RMS prediction differences from the dense network on 8192 uniform circle points, in legend order.

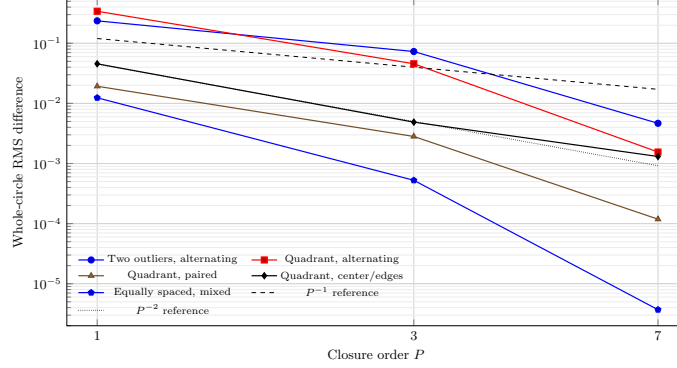


Figure 3: Whole-circle difference from dense training against order, two hidden tanh layers (left half of Table 2), with reference slopes. These are matched-loss discrepancies of fitted functions, not the same-time trajectory error bounded by Theorem 6.1.

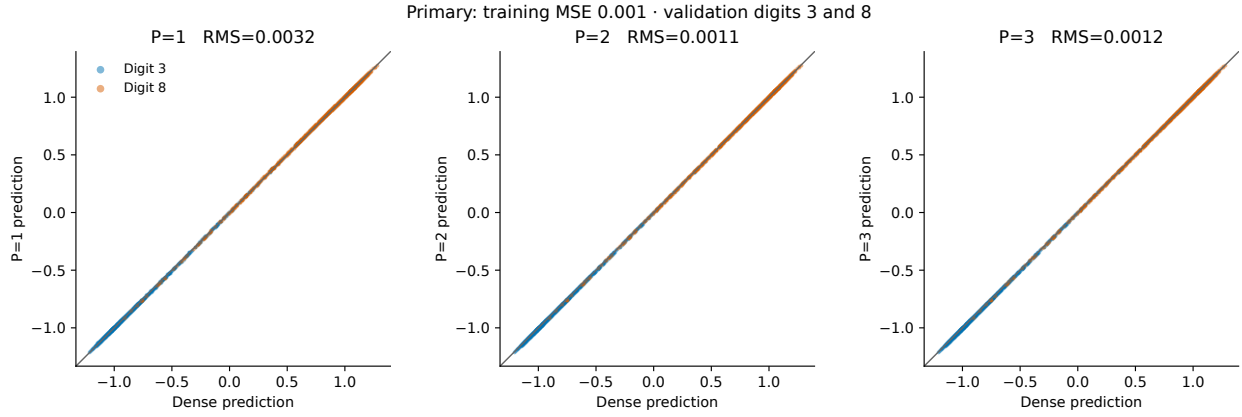


Figure 4: MNIST 3 versus 8, 100 training images, width 4096: response memory versus dense predictions on 1984 held-out images. RMS differences are 0.0032, 0.0011 and 0.0012 for $P = 1, 2, 3$.

82-configuration sweep, on the 65 configurations where dense and all orders fit, the joint clock's median differences were 0.031, 0.013 and 0.0066 for $P = 1, 2, 3$.

9 Discussion: toward an idealized model of learning

Compression provides a way to ask what a theory regards as the state of learning. A parameter description retains connections; a tangent hierarchy retains interactions among output derivatives; a dynamical field description retains response laws and correlations. These are different coordinates for asking how the present state determines subsequent learning. Their scientific value can be compared through the information retained, the accuracy supported, and the mechanisms preserved. Section 7 makes that comparison quantitative for specified representations, rather than assigning one universal numerical complexity to an entire framework.

Response memory identifies a constructive sufficient state for approximate training: summaries of how each neuron has responded forward and backward, coupled through the initialized network. These summaries keep changing with the features they help generate. At every fixed width, the convergence theorem controls this feedback over a complete finite interval and supplies a state—

accuracy trade-off. It establishes that the entire learned history need not be retained literally to reproduce deep nonlinear feature learning. The population version would replace the finite vectors by finitely many memory fields, as formulated in Conjecture 6.4. In that sense the construction connects the collective-description motivation of mean-field theory with a finite-order dynamical approximation.

The circle experiments test a consequence that training loss alone cannot resolve. Many functions interpolate eight labels; the radial comparisons show that a small memory state can also follow the fitted function selected away from those labels. Their matched-loss comparisons demonstrate usefulness on the tested tasks, while the same-time theorem explains why autonomous approximation is possible. The asymptotic constants do not yet explain the small empirical orders or imply a universal efficiency ranking.

The main remaining resources are visible in the representation. The fixed random environment W_0 is retained, and preserving its forward and adjoint reuse while compressing it is a further problem. Population error rates need width-uniform control, and longer horizons can require higher order even though the state formula contains no elapsed-step count. The theory covers smooth activations, and its memory grows with the number of training samples; a continuum of training inputs needs an additional representation of input dependence. These questions ask how far the same organizing principle can be pushed: which evolving information remains necessary at each desired accuracy?

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A Proof of the learning-speed clock theorem

Throughout the appendices we write $b_a^{(\ell)} = r_a \delta_a^{(\ell)} / \rho$ for the residual-weighted backward response, $\bar{h}_a^{(\ell-1)}$ and $\bar{b}_a^{(\ell)}$ for its and the forward response’s histories in the clock coordinate ξ (extended over the unit prefix as in the main text), and $h_a^{(\ell-1)*}$, $b_a^{(\ell)*}$ for the values at the current endpoint of their degree- $(P-1)$ Legendre projections. The only approximation-theoretic input is the following Legendre tail bound.

Lemma A.1 (Legendre tail). *For $q \in H^1(0, \tau; \mathbb{R}^N)$,*

$$\|q - \Pi_P q\|_{L^2(0, \tau)}^2 \leq \frac{1}{P(P+1)} \int_0^\tau \xi(\tau - \xi) \|q'(\xi)\|^2 d\xi \leq \frac{\tau^2}{4P(P+1)} \|q'\|_{L^2(0, \tau)}^2. \quad (13)$$

Proof. On $[0, 1]$, the shifted Legendre equation is

$$-\frac{d}{dx} \{x(1-x)p'_k(x)\} = k(k+1)p_k(x).$$

Integration by parts shows that the derivatives are orthogonal in the weight $x(1-x)$ and have squared norm $k(k+1)/(2k+1)$. Applying Bessel’s inequality to q' in this weighted space gives

$$\sum_{k \geq 1} k(k+1) \frac{\|c_k\|^2}{2k+1} \leq \int_0^1 x(1-x) \|q'(x)\|^2 dx,$$

where $c_k = (2k+1) \int_0^1 q(x)p_k(x) dx$. Parseval, polynomial density and $k(k+1) \geq P(P+1)$ for $k \geq P$ give the first bound on $[0, 1]$. Rescaling and $\xi(\tau - \xi) \leq \tau^2/4$ give (13); summing componentwise proves the vector statement. $\square$

Step 1: dense existence and a common compact region. The mobility matrix $\mathcal{D}$ has largest eigenvalue n , and $F = -\mathcal{D}\nabla\mathcal{L}$. Hence

$$\dot{\mathcal{L}} = -\left\|\mathcal{D}^{-1/2}\dot{\theta}\right\|^2, \quad \int_0^t \left\|\dot{\theta}(s)\right\|^2 ds \leq n\rho(0)^2. \quad (14)$$

Thus $\|\theta(t) - \theta(0)\| \leq \sqrt{nt}\rho(0)$. A finite maximal endpoint would have a finite limiting state, from which local existence would extend the solution, so the dense flow is global.

Now fix T and let

$$R = \|\theta(0)\| + \sqrt{nT}\rho(0) + 1, \quad \mathcal{B} = \{\vartheta : \|\vartheta\| \leq R\}.$$

The dense path stays at distance at least one from $\partial\mathcal{B}$. We stop the closure at its first exit from $\mathcal{B}$; the rest of the proof uses constants on this explicit ball and eventually rules out that exit.

To make the constants concrete, set $X = \max_a \|x_a\|/\sqrt{d}$ and $\alpha_0 = X$, and choose recursively

$$\begin{aligned} s_\ell &= \sup_{|z| \leq R\alpha_{\ell-1}} |\phi'(z)|, & \alpha_\ell &= \sqrt{n} \sup_{|z| \leq R\alpha_{\ell-1}} |\phi(z)|, \\ \beta_H &= Rs_H, & \beta_\ell &= s_\ell R\beta_{\ell+1} \quad (\ell < H). \end{aligned} \quad (15)$$

Then $\|h_a^{(\ell)}\| \leq \alpha_\ell$ and $\|\delta_a^{(\ell)}\| \leq \beta_\ell$ on $\mathcal{B}$. Put

$$\begin{aligned} Y &= \left(m^{-1} \sum_a y_a^2\right)^{1/2}, & q &= R\alpha_H/n + Y, & S &= Tq, & A &= 1 + S, \\ v_1 &= 2\beta_1 X, & v_\ell &= 2\beta_\ell \alpha_{\ell-1}/n, & v_w &= 2\alpha_H, \\ V &= \left(v_1^2 + \sum_{\ell=2}^H v_\ell^2 + v_w^2\right)^{1/2}. \end{aligned} \quad (16)$$

On the stopped interval, $\rho \leq q$, $\tau \leq A$ and $\|F\| \leq V\rho$. Since F is locally Lipschitz, it has some finite Lipschitz constant K on $\mathcal{B}$.

Step 2: an exact projection-error energy. For any single forward or backward history q , let $D_q(t) = \|q - \Pi_P q\|_{L^2(0,\tau(t))}^2$, and let q^* be the polynomial projection evaluated at the current endpoint. Differentiating the raw energy and the projected energy with (6), every basis-dilation term cancels, leaving

$$\dot{D}_q = \rho \|q - q^*\|^2, \quad D_q(t) = \int_0^t \rho(s) \|q(s) - q^*(s)\|^2 ds. \quad (17)$$

The jump at the end of the backward prefix is harmless, since only square integrability and growing integrals are used.

Next differentiate (7). The physical closure equation is exactly

$$\dot{\hat{\theta}}_P = F(\hat{\theta}_P) + E, \quad E_1 = E_w = 0, \quad (18)$$

with internal defects

$$E_\ell = \frac{2\rho}{nm} \sum_a (b_a^{(\ell)} - b_a^{(\ell)*})(h_a^{(\ell-1)} - h_a^{(\ell-1)*})^\top. \quad (19)$$

The two mixed projection terms vanish by orthogonality, so the defect is a pure product of errors. Consequently

$$\int_0^t \|E_\ell(s)\|_F \, ds \leq \frac{2}{nm} \sum_a \sqrt{D_{b,\ell,a}(t) D_{h,\ell,a}(t)}. \quad (20)$$

This inequality is the bridge from a same-history approximation bound to a trajectory theorem. Define also the proof-only accumulator

$$W_{\text{acc}}^{(\ell)}(t) = W_0^{(\ell)} - \frac{2}{nm} \sum_a \int_0^t \widehat{r}_a \widehat{\delta}_a^{(\ell)} (\widehat{h}_a^{(\ell-1)})^\top \, ds, \quad R_\ell = \widehat{W}^{(\ell)} - W_{\text{acc}}^{(\ell)}.$$

Then $R_\ell(0) = 0$ and $\dot{R}_\ell = E_\ell$, so (20) controls the accumulated absolute velocity error, not only a signed endpoint integral.

Step 3: forward histories have a P -independent derivative bound. Before changing variables, note that a nonstationary learning-speed solution cannot first reach $\rho = 0$ at a finite regular state. Every raw velocity vanishes there, and local uniqueness applied backward from that equilibrium would force the preceding path to be stationary. Hence $\xi = \tau(t)$ is a valid coordinate on every nonstationary compact segment; the raw algorithm itself remains defined at $\rho = 0$ without division.

Write primes for derivatives in $\xi = \tau(t)$ and define

$$Z_\ell(t) = \frac{1}{m} \sum_a \int_0^{\tau(t)} \left\| (h_a^{(\ell)})'(\xi) \right\|^2 \, d\xi.$$

The first layer satisfies $\left\| (h_a^{(1)})' \right\| \leq s_1 v_1 X$. For later layers, the chain rule gives

$$(h_a^{(\ell)})' = \phi'(z_a^{(\ell)}) \odot \left[(F_\ell/\rho + E_\ell/\rho) h_a^{(\ell-1)} + \widehat{W}^{(\ell)} (h_a^{(\ell-1)})' \right]. \quad (21)$$

The delicate term is the defect, because endpoint evaluation of a polynomial projection can amplify errors by a factor that grows with P . Endpoint evaluation of a degree- $(P-1)$ polynomial has operator norm $P/\sqrt{\tau}$ from $L^2(0, \tau)$, because $\sum_{k < P} (2k+1) = P^2$. Projection contraction and $m^{-1} \sum_a \left\| b_a^{(\ell)} \right\|^2 \leq \beta_\ell^2$ then imply

$$\left(m^{-1} \sum_a \left\| b_a^{(\ell)*} \right\|^2 \right)^{1/2} \leq P \beta_\ell, \quad \left(m^{-1} \sum_a \left\| b_a^{(\ell)} - b_a^{(\ell)*} \right\|^2 \right)^{1/2} \leq (P+1) \beta_\ell.$$

Combining this with Lemma A.1, (17) and (19) yields

$$\int_0^t \rho \|E_\ell/\rho\|_F^2 \, ds \leq \frac{2\beta_\ell^2 A^2}{n^2} Z_{\ell-1}(t). \quad (22)$$

Here the endpoint factor $P+1$ cancels the P^{-1} projection factor. This cancellation is the step that prevents a loss of convergence order with depth.

Define finite, P -independent bounds recursively by

$$\begin{aligned} \overline{Z}_1 &= S(s_1 v_1 X)^2, \\ \overline{Z}_\ell &= 3s_\ell^2 \left[S v_\ell^2 \alpha_{\ell-1}^2 + \left(R^2 + \frac{2\beta_\ell^2 A^2 \alpha_{\ell-1}^2}{n^2} \right) \overline{Z}_{\ell-1} \right]. \end{aligned} \quad (23)$$

Integrating (21), and using $\|x + y + z\|^2 \leq 3(\|x\|^2 + \|y\|^2 + \|z\|^2)$ together with (22), proves inductively that $Z_\ell \leq \overline{Z}_\ell$.

Step 4: accumulated defect and feedback. The zero backward prefix gives $m^{-1} \sum_a D_{b,\ell,a} \leq S\beta_\ell^2$, and Lemma A.1 gives

$$\frac{1}{m} \sum_a D_{h,\ell,a} \leq \frac{A^2 \bar{Z}_{\ell-1}}{4P(P+1)}.$$

Summing (20) over links, we obtain

$$\int_0^t \|E(s)\| \, ds \leq \frac{B_{\text{LS}}}{\sqrt{P(P+1)}}, \quad B_{\text{LS}} = \frac{A}{n} \sum_{\ell=2}^H \beta_\ell \sqrt{S \bar{Z}_{\ell-1}}. \quad (24)$$

It remains to account for feedback. Subtracting the dense flow from the perturbed flow and applying the Lipschitz bound on F gives

$$e(t) \leq \frac{B_{\text{LS}}}{\sqrt{P(P+1)}} + K \int_0^t e(s) \, ds,$$

and Gronwall's inequality yields

$$\sup_{t \leq T} e(t) \leq \frac{B_{\text{LS}} e^{KT}}{\sqrt{P(P+1)}}. \quad (25)$$

Choose P_0 so that the right side is less than $1/2$. The closure then cannot reach $\partial\mathcal{B}$. Its moment representations, $1 \leq \tau \leq A$ and the response bounds place its whole finite state in a compact subset of the locally Lipschitz domain. A finite maximal endpoint would therefore extend, so the closure exists through T . Lipschitz dependence of the network responses on θ gives the final prediction statement.

Globally bounded activations. Suppose now that every activation and its first derivative are globally bounded. Set

$$\alpha_\ell = \sqrt{n} \|\phi\|_\infty, \quad s_\ell = \|\phi'\|_\infty, \quad B_w = \sqrt{\|w(0)\|^2 + nY^2T}.$$

For both the dense and the closure flows, the unchanged readout equation obeys

$$\frac{d}{dt} \|w\|^2 = nY^2 - 4n \frac{1}{m} \sum_a (f_a - y_a/2)^2 \leq nY^2.$$

Therefore $\|w\| \leq B_w$, $\rho \leq q := B_w \alpha_H / n + Y$, $S = Tq$ and $\tau \leq A := 1 + S$. Start with $\beta_H = B_w s_H$ and proceed downward: for $\ell = H, \dots, 2$, define

$$D_\ell = \left\| W_0^{(\ell)} \right\|_F + \frac{2A\beta_\ell \alpha_{\ell-1}}{n}, \quad \beta_{\ell-1} = s_{\ell-1} D_\ell \beta_\ell.$$

Projection contraction in (8) gives $\left\| \widehat{W}^{(\ell)} \right\|_F \leq D_\ell$, and the backward recurrence then gives $\|\delta^{(\ell-1)}\| \leq \beta_{\ell-1}$. There is no circularity, because the descent starts from the bounded readout. Finally,

$$\left\| W^{(1)}(t) \right\|_F \leq \left\| W^{(1)}(0) \right\|_F + 2SX\beta_1.$$

These P -independent physical bounds, the moment integral representations and $1 \leq \tau \leq A$ place the order- P state in a compact set for every fixed P , so the closure continues through T for every $P \geq 1$. Repeating Steps 2–4 with these global constants gives (25) for every P . This completes the proof. $\square$

B The joint feature-learning clock: construction and proof

Construction. Let Ψ be the concatenation over $\ell \geq 2$ and a of the forward and residual-weighted backward responses $(h_a^{(\ell-1)}, b_a^{(\ell)})$, so that the clock (9) reads $\dot{\tau} = g = \rho + \|\dot{\Psi}\|_2$ and $\|\mathrm{d}\Psi/\mathrm{d}\tau\| \leq 1$. History is placed at $\xi = \tau(s)$ with mass $\rho(s) \mathrm{d}s$, while the prefix $[0, 1]$ carries Lebesgue mass and constant initial responses; the resulting measure μ_t satisfies $\mathrm{d}\mu_t \leq \mathrm{d}\xi$. With $p = (p_0, \dots, p_{P-1})^\top$ evaluated at ξ/τ , $e = (1, \dots, 1)^\top$, and the triangular matrix $\mathbf{T}_{kk} = k$, $\mathbf{T}_{kj} = 2j + 1$ for $j < k$ (zero above the diagonal), the state consists of the moment matrices $\bar{h}_a^{(\ell-1)} = \int \bar{h}_a^{(\ell-1)}(\xi) p^\top \mathrm{d}\mu_t$, $\bar{\delta}_a^{(\ell)} = \int \bar{b}_a^{(\ell)}(\xi) p^\top \mathrm{d}\mu_t$ and one shared Gram matrix $G = \int pp^\top \mathrm{d}\mu_t$, which evolve by

$$\begin{aligned} \widehat{W}^{(\ell)} &= W_0^{(\ell)} - \frac{2}{nm} \sum_a (\bar{\delta}_a^{(\ell)} G^{-1} (\bar{h}_a^{(\ell-1)})^\top - b_a^{(\ell)}(0) h_a^{(\ell-1)}(0)^\top), \\ \dot{\bar{h}}_a^{(\ell-1)} &= \rho h_a^{(\ell-1)} e^\top - (g/\tau) \bar{h}_a^{(\ell-1)} \mathbf{T}^\top, \quad \dot{\bar{\delta}}_a^{(\ell)} = r_a \delta_a^{(\ell)} e^\top - (g/\tau) \bar{\delta}_a^{(\ell)} \mathbf{T}^\top, \\ \dot{G} &= \rho e e^\top - (g/\tau) (\mathbf{T}G + G\mathbf{T}^\top), \end{aligned} \tag{26}$$

with $G(0) = \text{diag}(1, \frac{1}{3}, \dots, \frac{1}{2P-1})$, only the zeroth moment columns nonzero initially, and the prefix product subtracted so that $\widehat{W}^{(\ell)}(0) = W_0^{(\ell)}$. With $h_a^{(\ell-1)*} = \bar{h}_a^{(\ell-1)} G^{-1} e$ and $b_a^{(\ell)*} = \bar{\delta}_a^{(\ell)} G^{-1} e$, differentiating the reconstruction cancels every g/τ term and gives the defect $E_\ell = \frac{2\rho}{nm} \sum_a (b_a^{(\ell)} - b_a^{(\ell)*})(h_a^{(\ell-1)} - h_a^{(\ell-1)*})^\top$, of the same product form as (19).

Step 1: a lower bound on the dense residual. Assume $\rho_0 > 0$ as stated, and use the compact physical ball $\mathcal{B}$ from Appendix A. Retain the constants $\alpha_\ell, \beta_\ell, q, S, \mathbf{A}, V, K$ from the proof of Theorem 6.1. On this ball the output map has a finite Lipschitz constant c_f : prediction and residual RMS change by at most $c_f \|\theta - \theta'\|$. Along dense flow,

$$|\dot{\rho}_{\text{dense}}| \leq c_f \|F(\theta)\| \leq c_f V \rho_{\text{dense}}.$$

Consequently

$$\rho_{\text{dense}}(t) \geq \mu := \rho_0 e^{-c_f V T} > 0, \quad 0 \leq t \leq T. \tag{27}$$

Step 2: an exact weighted projection energy. Consider the maximal regular closure interval stopped before either its physical state leaves $\mathcal{B}$ or its residual reaches $\mu/2$. We first derive a clock bound on this stopped interval without assuming small projection error. For any inserted vector history f , let

$$D_f = \|f - \Pi_t f\|_{L^2(\mu_t)}^2 = \int \|f\|^2 \mathrm{d}\mu_t - \text{tr}(B_f G^{-1} B_f^\top).$$

Differentiating $G^{-1}G = I$ gives

$$\frac{\mathrm{d}}{\mathrm{d}t} G^{-1} = -\rho G^{-1} e e^\top G^{-1} + (g/\tau)(G^{-1} \mathbf{T} + \mathbf{T}^\top G^{-1}).$$

Together with (26), this cancels every dilation term and yields

$$\dot{D}_f = \rho \|f - f^*\|^2, \quad D_f(t) = \int_0^t \rho \|f - f^*\|^2 \mathrm{d}s. \tag{28}$$

The initial error is zero because every prefix is constant. Thus (20) holds with the weighted errors in (28).

Step 3: order-independent coarse bounds. Put

$$Q = m \sum_{\ell=2}^H (\alpha_{\ell-1}^2 + \beta_{\ell}^2), \quad C_0 = \frac{AQ}{nm}.$$

The total mass of μ_t is at most $A = 1 + S$. Projection error is no larger than raw history energy, including the matching prefixes. Applying $2\sqrt{uv} \leq u + v$ in the defect estimate gives the order-independent bounds

$$\int_0^t \|E\| \, ds \leq C_0, \quad \int_0^t \|\dot{\hat{\theta}}\| \, ds \leq VS + C_0. \quad (29)$$

The second inequality uses $\dot{\hat{\theta}} = F(\hat{\theta}) + E$ and $\int_0^t \|F\| \, ds \leq V \int_0^t \rho \, ds \leq VS$. This total-variation estimate is stronger than bounded physical position, and it is obtained before any clock bound.

Step 4: a clock bound without a cap. It remains to control the derivative of the normalized response map. Let $q_r = r/\rho \in \mathbb{R}^m$. Then

$$\|q_r\|^2 = m, \quad Dq_r[v] = \left(I - \frac{q_r q_r^\top}{m} \right) \frac{Dr[v]}{\rho}. \quad (30)$$

The matrix in parentheses is an orthogonal projector, and $\rho \geq \mu/2$ on the stopped interval. Differentiating the forward and backward recurrences uses only bounded physical weights and bounded first two activation derivatives on compact preactivation intervals; the first layer uses only ϕ' . Hence there is a finite, P -independent constant J such that

$$\|D\Psi(\vartheta)\|_{\text{op}} \leq J$$

throughout the stopped tube. The stated local smoothness also makes this derivative locally Lipschitz. Since $\dot{\Psi} = D\Psi(\hat{\theta})\hat{\theta}$, (29) now gives

$$\tau_P(t) = 1 + \int_0^t (\rho + \|\dot{\Psi}\|) \, ds \leq A + J(VS + C_0) =: \Lambda_T. \quad (31)$$

No clock cap has been assumed or inserted.

Step 5: the sharp joint approximation estimate. We can now use the sharper approximation estimate. The entire concatenated history $\bar{\Psi}(\xi)$ is constant on the prefix, continuous at its join and, on the physical history,

$$\|\partial_\xi \bar{\Psi}\| = \frac{\|\dot{\Psi}\|}{\rho + \|\dot{\Psi}\|} \leq 1.$$

Therefore $\int_0^{\tau_P} \|\bar{\Psi}'\|^2 \, d\xi \leq \tau_P - 1$. The weighted projection is a best approximation in $L^2(\mu_t)$, and $d\mu_t \leq d\xi$, so comparison with the ordinary Legendre projection and Lemma A.1 give the joint bound

$$\sum_{\ell,a} (D_{h,\ell,a} + D_{b,\ell,a}) \leq \frac{\tau_P^2(\tau_P - 1)}{4P(P+1)} \leq \frac{\Lambda_T^2(\Lambda_T - 1)}{4P(P+1)}. \quad (32)$$

This is one estimate for the concatenated path; it does not introduce a separate dimension factor per unit or layer.

Step 6: feedback and continuation. Combining (20), (31), (32) and $2\sqrt{uv} \leq u + v$ yields

$$\int_0^t \|E\| \, ds \leq \frac{\Gamma_{\text{JF}}}{P(P+1)}, \quad \Gamma_{\text{JF}} = \frac{\Lambda_T^2(\Lambda_T - 1)}{4nm}. \quad (33)$$

Subtracting the dense flow and applying the Lipschitz constant K and Gronwall's inequality gives

$$\left\| \hat{\theta}_P(t) - \theta(t) \right\| \leq \frac{\Gamma_{\text{JF}} e^{KT}}{P(P+1)}. \quad (34)$$

Choose $P_0(T)$ so that, for every $P \geq P_0(T)$,

$$\frac{\Gamma_{\text{JF}} e^{KT}}{P(P+1)} \leq \frac{1}{2}, \quad c_f \frac{\Gamma_{\text{JF}} e^{KT}}{P(P+1)} \leq \frac{\mu}{4}. \quad (35)$$

The first inequality rules out physical exit, because the dense path stays at distance one from $\partial\mathcal{B}$. The second, together with (27), gives $\hat{\rho} \geq 3\mu/4 > \mu/2$, ruling out residual exit.

Finally, the matching prefix supplies, at every fixed P ,

$$G(t) \succeq \int_0^1 p(\xi/\tau_P(t)) p(\xi/\tau_P(t))^\top \, d\xi. \quad (36)$$

The right side is positive definite, since a zero quadratic form would make a nonzero polynomial vanish on an interval. Its entries vary continuously with τ_P , so on $1 \leq \tau_P \leq \Lambda_T$ its least eigenvalue has a positive minimum $\gamma_{P,T}$; this minimum need not be uniform in P . The moment integral representations, finite history mass, bounded responses and bounded fixed-order polynomials then bound every raw coordinate. Together with the physical, residual, clock and Gram bounds, the full state lies in a compact subset of its locally Lipschitz domain. A finite maximal endpoint therefore extends, which proves continuation through T and completes the theorem with $C_T = \Gamma_{\text{JF}}$ and $K_T = K$. $\square$

Implementation of the joint clock

Equation (9) appears implicit only because $\dot{\Psi}$ is written compactly; no nonlinear solve for g is needed. The internal velocities do not depend on g , so they can be computed first. Begin with $q = G^{-1}e$, the endpoint fits a^*, b^* , and the internal velocities

$$V_\ell = \dot{\widehat{W}}^{(\ell)} = -\frac{2\rho}{nm} \sum_a \left[b_a^{(\ell)} (h_a^{(\ell-1)*})^\top + b_a^{(\ell)*} (h_a^{(\ell-1)} - h_a^{(\ell-1)*})^\top \right]. \quad (37)$$

These expressions contain no g . Next compute $V_1 = F_1$ and $V_w = F_w$, and run a directional forward pass,

$$\dot{z}_a^{(1)} = V_1 x_a / \sqrt{d}, \quad \dot{z}_a^{(\ell)} = V_\ell h_a^{(\ell-1)} + \widehat{W}^{(\ell)} \dot{h}_a^{(\ell-1)}, \quad \dot{h}_a^{(\ell)} = \phi'(z_a^{(\ell)}) \odot \dot{z}_a^{(\ell)},$$

which gives $\dot{r}$ and $\dot{\rho}$. A reverse directional pass then differentiates the backward recurrence, using ϕ'' for $\ell \geq 2$, and yields

$$\dot{h}_a^{(\ell)} = \frac{\dot{r}_a \delta_a^{(\ell)} + r_a \dot{\delta}_a^{(\ell)}}{\rho} - b_a^{(\ell)} \frac{\dot{\rho}}{\rho}.$$

Only then do we evaluate $g = \rho + \left\| \dot{\Psi} \right\|$ and, finally, (26). Matrix and transpose actions use the same reconstructed operator. In practice one factors G and solves linear systems; an explicit inverse is unnecessary.

The joint clock is more regular in approximation coordinates, but it is not automatically better at low order. At $P = 1$ the polynomial space consists of constants, so changing the coordinate alone leaves the physical constant-mode closure unchanged. Higher orders pay for Gram solves and derivative passes, and the theoretical constants can be large.

C A frozen-span closure of the population flow

This appendix records a closure of the population flow itself, for the smallest case allowed by the contract: two hidden layers, $\phi = \tanh$, inputs on a circle ($d = 2$, $x/\sqrt{d} \in S^1$) with labels $|y| \leq Y$. The population state is the first-layer weight row $W^{(1)} \in L^2(\Omega_1; \mathbb{R}^2)$, the readout $w \in L^2(\Omega_2)$ and the middle operator $W^{(2)} = W_0^{(2)} + (W^{(2)} - W_0^{(2)})$, where $W_0^{(2)}$ is the initialized Gaussian operator and the learned part is Hilbert–Schmidt. The flow is (2) with $\frac{1}{m} \sum_a$ replaced by integration against a training law μ on inputs and labels, and the reverse action is always the true adjoint $(W^{(2)})^*$. We work on $0 \leq t \leq T = 1/200$ and on a Wasserstein neighborhood of the two-point task $\nu_* = \frac{1}{2}\delta_{(\sqrt{2}e_1, +1)} + \frac{1}{2}\delta_{(\sqrt{2}e_2, -1)}$; the neighborhood contains perturbed, nonorthogonal, nonatomic and noisy input laws, and neither its radius nor the interval shrinks with the closure order.

The compressed state. Enumerate the finite expressions built from constants, the Gaussian initial weights, rational affine combinations, $\sin$, $\cos$, ϕ , bounded products, and the initialized actions $W_0^{(2)}$ and $(W_0^{(2)})^*$. At order N this gives, on each population, a finite vector of *observables of the initialization*, which we normalize by a ridge-regularized Gram matrix so that they have unit scale. The saved state is then

- the joint law, on population 1, of the current first-layer weights $W^{(1)}$, their initial values, and the N observables;
- the joint law, on population 2, of the current readout w and the N observables;
- one finite matrix M representing the middle-layer action $W^{(2)}$ between the two observable vectors, initialized from $W_0^{(2)}$.

The closure evolves this state by the population flow itself, with $W^{(2)}$ replaced by the finite operator “project onto population-1 observables, apply M , expand in population-2 observables” and $(W^{(2)})^*$ by its transpose (details below). It is autonomous, uses only the current state and the training law, and is restartable from its own saved state. Initial and current values are stored jointly, so same-neuron quantities such as feature displacement are genuine observables of the state.

Theorem C.1 (Autonomous population closure). *For every separately fixed training law in the stated neighborhood of ν_* and every order N , the closure is uniquely well posed on $[0, T]$ and restartable from its saved state, and as $N \rightarrow \infty$ its prediction f_N converges to the population prediction f uniformly on $[0, T]$ and over the whole input circle. Every separately fixed same-population joint observation—initial and current hidden pairs, feature displacements, forward and adjoint actions, bounded nonlinear pushforwards and quadratic contractions—converges uniformly on $[0, T]$ in Wasserstein distance W_2 .*

The proof (below) controls the distance between closure and population flow through the error of the finite observable basis; unbounded backward fields force a truncation that turns Gronwall’s inequality into an Osgood inequality, whose solution still tends to zero with the basis error.

Proposition C.2 (Both layers learn, nonlinearly). *On a Wasserstein neighborhood of ν_* there is a common time at which both hidden populations have strictly positive mean-square displacement of their activations from initialization, and strictly positive nonaffinity $\inf_{\alpha, \beta} \mathbb{E}|\phi(z) - \alpha z - \beta|^2$ of their preactivations.*

The small-time expansion of the flow at ν_* (below) shows that both hidden displacements are nonzero at order t^2 , a Gaussian preactivation has positive nonaffinity, and continuity extends both properties to a neighborhood. The closure therefore converges on a family where all three properties of the introduction hold.

What this compresses, and why it is not the main result. The learned operator $W^{(2)} - W_0^{(2)}$ of the population flow is replaced by a finite matrix between two populations: an operator ODE becomes a population ODE plus a finite matrix. But the matrix acts between *frozen* spans of initialization observables, so the learned middle layer can only rotate within a span fixed at initialization, while $W^{(1)}$ and w move freely. This is a frozen-dictionary closure of the middle layer, and it shows the characteristic limitations of that class: convergence comes only from the spans becoming dense, there is no rate in N , and the guarantee holds on a short horizon near a reference task. The two laws are continuum objects, so the state is not a finite list of scalars. The frozen-dictionary experiments of Section 8 show the same limitation at finite width, and the response memories of Section 6, whose atoms coevolve with the features, are the answer to it.

Details

Population flow. With $\phi = \tanh$, $d = 2$ and inputs $x \in \sqrt{2}S^1$, the population fields are $z^{(1)}(x) = W^{(1)} \cdot x / \sqrt{d}$, $h^{(1)} = \phi(z^{(1)})$, $z^{(2)}(x) = W^{(2)}h^{(1)}(x)$, $h^{(2)} = \phi(z^{(2)})$, $\delta^{(2)}(x) = w\phi'(z^{(2)}(x))$, $\delta^{(1)}(x) = \phi'(z^{(1)}(x))(W^{(2)})^*\delta^{(2)}(x)$ and $f(x) = \mathbb{E}_2[w h^{(2)}(x)]$; with $r = f - y$ and a training law μ ,

$$\dot{W}^{(1)} = -2 \int r \delta^{(1)} \frac{x}{\sqrt{d}} d\mu, \quad \dot{W}^{(2)} = -2 \int r \delta^{(2)} \otimes h^{(1)} d\mu, \quad \dot{w} = -2 \int r h^{(2)} d\mu.$$

Observables. Let $\psi_1 \in \mathbb{R}^{N_1}$ and $\psi_2 \in \mathbb{R}^{N_2}$ be the order- N vectors of initialization observables on the two populations, ridge-normalized by $\psi_\ell \mapsto (\mathbb{E}_\ell[\psi_\ell \psi_\ell^\top] + 2^{-N}I)^{-1/2} \psi_\ell$, so that the projections $g \mapsto \psi_\ell^\top \mathbb{E}_\ell[\psi_\ell g]$ are contractions converging strongly to the identity on the observable spaces. The initialized middle action is recorded in $M_0 = \mathbb{E}_2[\psi_2 (W_0^{(2)} \psi_1)^\top]$; the forward and adjoint actions of $W_0^{(2)}$ on observables are drawn from one correlated Gaussian source, which is what makes the two orientations one operator and its adjoint.

Closure. The state is $\text{Law}(\psi_1, W_0^{(1)}, W^{(1)})$ on population 1, $\text{Law}(\psi_2, w)$ on population 2, and $M \in \mathbb{R}^{N_2 \times N_1}$, initialized with $W^{(1)} = W_0^{(1)}$, $w = 0$ and $M = M_0$. The closure is the population flow above with $W^{(2)}$ replaced by

$$W_N^{(2)} g = \psi_2^\top M \mathbb{E}_1[\psi_1 g], \quad (W_N^{(2)})^* g = \psi_1^\top M^\top \mathbb{E}_2[\psi_2 g],$$

$$\dot{M} = -2 \int r_N \mathbb{E}_2[\psi_2 \delta^{(2)}] \mathbb{E}_1[\psi_1 h^{(1)}]^\top d\mu,$$

so that $z_N^{(2)}(x) = \psi_2^\top M \mathbb{E}_1[\psi_1 h^{(1)}(x)]$ and $f_N(x) = \mathbb{E}_2[w h_N^{(2)}(x)]$; the weights $W^{(1)}$ and w move by the flow equations with r_N and $\delta_N^{(1)}$, which transports the two laws. The represented operator is $W_N^{(2)}$, whose learned part is $M - M_0$ written between observable spaces.

Well-posedness. For fixed N the characteristic equations are locally Lipschitz. The loss obeys $\frac{d}{dt} \mathcal{L}_N = -\left\| \dot{W}^{(1)} \right\|_2^2 - \left\| \dot{M} \right\|_F^2 - \left\| \dot{w} \right\|_2^2$, and $\mathcal{L}_N(0) \leq Y^2$ gives $\|w(t)\|_\infty \leq 2Yt$, $\|M(t) - M_0\|_F \leq 2Y^2 t^2$, $\left\| W_N^{(2)}(t) \right\|_{\text{op}} \leq 2 + 2Y^2 t^2$ and $\|W^{(1)}(t)\|_2 \leq \sqrt{2} + 4Y^2 t^2 + 2Y^4 t^4$, which exclude finite-time escape. The joint laws contain the correlations between observables and current weights, so the saved state suffices to restart.

Convergence. Place closure and population flow on common probability spaces and set

$$e_N = \left\| W_N^{(1)} - W^{(1)} \right\|_2 + \left\| (W_N^{(2)} - W_{N,0}^{(2)}) - (W^{(2)} - W_0^{(2)}) \right\|_{\text{HS}} + \|w_N - w\|_2,$$

where $W_{N,0}^{(2)}$ is the closure's initial operator. The basis error

$$\begin{aligned} \varepsilon_N = & \sup_{t,x} \left\| (W_{N,0}^{(2)} - W_0^{(2)})h^{(1)}(t,x) \right\|_2 + \sup_{t,x} \left\| (W_{N,0}^{(2)} - W_0^{(2)})^* \delta^{(2)}(t,x) \right\|_2 \\ & + \sup_t \left\| \Pi_2 \dot{W}^{(2)}(t) \Pi_1 - \dot{W}^{(2)}(t) \right\|_{\text{HS}}, \end{aligned}$$

with Π_ℓ the observable projections, tends to zero by strong convergence of the projections, compactness of the population response curves in (t, x) , and density of finite-rank operators. Forward subtraction gives, uniformly in x , $\left\| h_N^{(1)} - h^{(1)} \right\|_2 \leq e_N$ and $\left\| h_N^{(2)} - h^{(2)} \right\|_2 + |f_N - f| \leq C(e_N + \varepsilon_N)$. Backward subtraction multiplies a changed gate by an unbounded reference field; with the exponential tail bound of the reference trajectory, truncating at level R gives

$$e'_N \leq C(1 + R)(e_N + \varepsilon_N) + C e^{-aR}.$$

Choosing $R = 1 + a^{-1} \log(1/v)$ with $v = e_N + \varepsilon_N + \eta$ yields $v' \leq Lv \log(e/v)$, hence

$$e_N(t) + \varepsilon_N + \eta \leq e^{1-\alpha(t)}(\varepsilon_N + \eta)^{\alpha(t)}, \quad \alpha(t) = e^{-Lt}.$$

Letting $\eta \downarrow 0$ and $N \rightarrow \infty$ gives $\sup_{t \leq T} e_N \rightarrow 0$, and the forward bounds give uniform convergence of f_N on $[0, T] \times S^1$. The same induction over observation programs controls every admissible joint observation in L^2 , and the canonical coupling bounds its W_2 distance by the L^2 distance.

Feature learning at ν_* . Write $g_a = W_0^{(1)} \cdot e_a$ for the initial preactivations at the two inputs, $h_a = \phi(g_a)$, $\xi_a = W_0^{(2)} h_a$ and $S = \phi(\xi_1) - \phi(\xi_2)$. With $U_a = S \phi'(\xi_a)$, $P_a = (W_0^{(2)})^* U_a$ and $V_a = \phi'(g_a)^2 P_a$, the population flow satisfies $w(t) = tS + o(t)$, $h^{(1)}(t, e_a) - h_a = \frac{y_a}{2} t^2 V_a + o(t^2)$ and $h^{(2)}(t, e_a) - \phi(\xi_a) = \frac{y_a}{2} t^2 \phi'(\xi_a) (\mathbb{E}[\phi^2(G)] U_a + W_0^{(2)} V_a) + o(t^2)$ for a standard Gaussian G . Adjunction gives $\mathbb{E}[h_a P_a] = y_a \mathbb{E}[\xi_a \phi(\xi_a) \phi'(\xi_a)] \neq 0$, so both hidden displacements are nonzero at order t^2 ; a nondegenerate Gaussian preactivation has positive nonaffinity since $\tanh$ is not affine on any interval; and continuity of the flow in the training law extends both properties to a neighborhood of ν_* .

D Exact compression in a deep linear benchmark

Take one datum $x = 1$, a label y , three linear hidden layers with order-one initialization, mobilities $(n\kappa, \kappa, \kappa, n\kappa)$, and the normalized factors $u = W^{(1)}/\sqrt{n}$, $v = W^{(4)}/\sqrt{n}$, so that $f_n = v^\top W^{(3)} W^{(2)} u$. Packing the four factors into the cyclic block matrix $\mathcal{C}_n$ with blocks $v^\top, u, W^{(2)}, W^{(3)}$ below and above the diagonal gives $f_n = \frac{1}{4} \text{Tr} \mathcal{C}_n^4$, and the feature derivative of $\mathcal{C}_n$ is $(\mathcal{C}_n^\top)^3$.

Theorem D.1 (Exact operator closure). *There are a separable Hilbert space, a bounded initialized source $\mathcal{C}_0$ and a unique global trace-class increment $Q(t)$ with*

$$\mathcal{C} = \mathcal{C}_0 + Q, \quad f = \frac{1}{4} \text{Tr} \mathcal{C}^4, \quad \dot{Q} = -2\kappa(f - y)(\mathcal{C}_0^* + Q^*)^3, \quad Q(0) = 0.$$

The kernel is $K = \text{Tr}[\mathcal{C}^3(\mathcal{C}^)^3]$, the residual obeys $\dot{r} = -2\kappa r K$, and $\|Q(t)\|_1 \leq 2|y|\sqrt{\kappa t}$. For every fixed T , finite gradient flow, and gradient descent for every step sequence tending to zero, converge in probability to $(f, K, \mathcal{L})$ uniformly on $[0, T]$.*

The proof builds $\mathcal{C}_0$ as the limit of initialized word statistics on a colored Fock space, obtains global existence from trace-norm control, passes to the width limit through finite-word Picard iterates, and treats gradient descent as an exact Euler step for Q .

Theorem D.2 (Restricted scalar nonclosure). *In the same model, no encoder formed from a fixed finite list of complete tensor contractions of $(u, W^{(2)}, W^{(3)}, v)$ with connected-graph degree bounded independently of width admits a width-independent polynomial autonomous ODE (or finite-order local polynomial PDE) and a polynomial output readout reproducing the output dynamics on a nonempty open set of states for all sufficiently large widths.*

The $(2j - 1)$ st feature derivative of the output contains the connected contraction

$$u^\top ((W^{(2)})^\top (W^{(3)})^\top W^{(3)} W^{(2)})^j u$$

of degree $4j + 2$. Polynomial operations on a bounded-degree alphabet form only disjoint unions of connected components already present, and nonisomorphic contraction graphs are linearly independent at large width, so this term cannot be synthesized. The theorem does not exclude arbitrary encoders, unrestricted fields, or closures tailored to a single initialized trajectory.

E Accuracy and resource comparisons: derivations

We compare approximations to the same population predictor f_∞ of (2), assumed to exist on $[0, T]$, with the same data and initialization law, including the readout. Matching initial predictions alone does not match the initial tangent kernel. Throughout, constants can depend on m, H, T , the data, initialization and test law; displayed factors in these variables do not exhaust that dependence. Population error estimates below are stated hypotheses, whereas the fixed-width response-memory results are Theorems 6.1 and 6.2. We suppress the T subscripts on the error constants of Section 7.

E.1 Test error, allocations and arithmetic

For a test probability measure Q , define

$$E_Q(g; T) = \sup_{t \leq T} \|g(t) - f_\infty(t)\|_{L^2(Q)}, \quad \mathcal{E}_Q(g; T) = (\mathbb{E} E_Q(g; T)^2)^{1/2}. \quad (38)$$

The reverse triangle inequality bounds the difference between the test RMSEs of g and f_∞ , against any $y_* \in L^2(Q)$, by $E_Q(g; T)$. This measures approximation, without assuming that f_∞ generalizes well. Moreover, pointwise estimates $\mathbb{E} \sup_{t \leq T} |f_n(t, x) - f_\infty(t, x)|^2 \leq C_T(x)^2/n$ with $C_T \in L^2(Q)$ give $\mathcal{E}_Q(f_n; T)^2 \leq n^{-1} \int C_T^2 dQ$ by moving the time supremum inside the integral and applying Tonelli. An input supremum would require additional spatial control.

For $\phi = \tanh$, set $a_\ell = \|h^{(\ell)} - h'^{(\ell)}\|_2 / \sqrt{n}$ and assume $\|x\| / \sqrt{d} \leq X$. Boundedness and Lipschitz continuity of $\tanh$ give

$$a_1 \leq X \frac{\|W^{(1)} - W'^{(1)}\|_F}{\sqrt{n}}, \quad a_\ell \leq \|W^{(\ell)} - W'^{(\ell)}\|_{\text{op}} + \|W'^{(\ell)}\|_{\text{op}} a_{\ell-1},$$

$$|f_n - f'_n| \leq \frac{\|w - w'\|_2}{\sqrt{n}} + \frac{\|w'\|_2}{\sqrt{n}} a_H.$$

Induction transfers parameter control to every such test input when operator and normalized readout norms are bounded. Coupling dense and compressed networks through their initialization, the

triangle inequality and Minkowski give

$$\mathcal{E}_Q(\hat{f}_{n,P}; T) \leq \mathcal{E}_Q(f_n; T) + \left(\mathbb{E} \sup_{t \leq T} \left\| \hat{f}_{n,P}(t) - f_n(t) \right\|_{L^2(Q)}^2 \right)^{1/2}. \quad (39)$$

Thus the conditional model $An^{-1/2} + B_q P^{-q}$, $q = 1, 2$, needs both a matched root-width estimate and closure constants and sufficient-order thresholds uniform in width, with appropriate moments. The perturbative fluctuation calculation of Bordelon and Pehlevan (2023) motivates the first estimate but is not a general remainder theorem for this model.

Moving memory counts persistent updated scalars; fixed matrices, coefficient evaluators and workspace are reported separately. The leading response-memory count is $2(H-1)mnP$. With $u = A/\sqrt{n}$, $v = B_q/P^q$ and $u + v = \varepsilon$, its minimization reduces to minimizing $u^{-2}(\varepsilon - u)^{-1/q}$. The logarithm has derivative $-2/u + 1/[q(\varepsilon - u)]$ and positive second derivative, and diverges at both endpoints. Hence the unique continuous optimum is

$$u_* = \frac{2q\varepsilon}{2q+1}, \quad v_* = \frac{\varepsilon}{2q+1}, \quad n_* = \frac{(2q+1)^2 A^2}{4q^2 \varepsilon^2}, \quad P_* = \left(\frac{(2q+1)B_q}{\varepsilon} \right)^{1/q}. \quad (40)$$

For $q = 1$, $(n_*, P_*) = (9A^2/(4\varepsilon^2), 3B_1/\varepsilon)$ and $M_* \sim \frac{27}{2}(H-1)mA^2B_1\varepsilon^{-3}$. For $q = 2$, $(n_*, P_*) = (25A^2/(16\varepsilon^2), \sqrt{5}B_2/\varepsilon)$ and $M_* \sim \frac{25\sqrt{5}}{8}(H-1)mA^2\sqrt{B_2}\varepsilon^{-5/2}$. Upward rounding is feasible. Outer weights, $(d+1)n$, and the joint Gram matrix, $O(P^2)$, perturb the finite-tolerance optimum but are lower order. Dense training instead takes $n = \lceil (A/\varepsilon)^2 \rceil$ and $M_* \sim (H-1)A^4\varepsilon^{-4}$. These optimize sufficient bounds in specified representations, not actual errors or all possible algorithms.

For completeness, full-batch arithmetic per right-hand-side evaluation is

$$\begin{aligned} C_{\text{dense}} &= O(Hmn^2 + mnd), \\ C_{\text{old}} &= O(Hmn^2 + Hm^2nP + HmnP + mnd), \\ C_{\text{joint}} &= O(Hmn^2 + Hm^2nP + HmnP^2 + P^3 + mnd). \end{aligned} \quad (41)$$

The terms respectively account for dense initialized actions, learned factors, moment updates or Gram solves, and Gram factorization. Directional passes for the joint clock add a constant number of network passes. Both closures retain $(H-1)n^2$ fixed entries; activation workspace is $O(Hmn)$, and Gram conditioning can affect numerical step counts. Consequently neither their total-storage nor their arithmetic exponent improves automatically.

E.2 A matched tangent hierarchy

With $\mathcal{D}_n = \text{diag}(nI, I, \dots, I, nI)$, define recursively

$$\begin{aligned} K^{(2)}(x, z) &= \nabla_\theta f(x)^\top \mathcal{D}_n \nabla_\theta f(z), \\ K^{(j+1)}(x_1, \dots, x_{j+1}) &= (\nabla_\theta K^{(j)}(x_1, \dots, x_j))^\top \mathcal{D}_n \nabla_\theta f(x_{j+1}). \end{aligned}$$

The chain rule gives the exact, mobility-matched hierarchy

$$\dot{f}(x) = -\frac{2}{m} \sum_a r_a K^{(2)}(x, x_a), \quad \dot{K}^{(j)} = -\frac{2}{m} \sum_a r_a K^{(j+1)}(\dots, x_a). \quad (42)$$

Truncation through order $p \geq 2$ retains exact initial coefficients and freezes $K^{(p)}$. A population version additionally requires limiting coefficients and a valid limiting hierarchy.

Feedback bound. Suppose $|K^{(j)}| \leq B_j$ along the exact flow, uniformly when the first argument is a training or test input and the others are training inputs. Let $R = 1 + \sup_{t,a} |r_a(t)|$ and stop the truncated system at predictor error one. Its residuals are then bounded by R . For supremum predictor error e_f and order- j kernel error e_j , subtraction, splitting each product using the exact kernel and truncated residual, yields

$$\begin{aligned} e_p(t) &\leq 2RB_{p+1}t, \\ e_j(t) &\leq \int_0^t (2Re_{j+1} + 2B_{j+1}e_f) ds \quad (2 \leq j < p), \\ e_f(t) &\leq \int_0^t (2Re_2 + 2B_2e_f) ds. \end{aligned} \quad (43)$$

Successive substitution integrates the top forcing p times. Each feedback term has convolution kernel $2B_{k+2}(2R(t-s))^k/k!$, $0 \leq k \leq p-2$. Bounding these kernels at T and applying Gronwall gives

$$\sup_{t \leq T} e_f(t) \leq \eta_p(T) := B_{p+1} \frac{(2RT)^p}{p!} \exp(L_{p,T}T), \quad L_{p,T} = 2 \sum_{k=0}^{p-2} B_{k+2} \frac{(2RT)^k}{k!}. \quad (44)$$

If $\eta_p(T) < 1$, first exit is impossible. The triangular integral equations then bound all training tensors, preventing finite-time escape and extending the solution through T . The test bound implies $E_Q \leq \eta_p$ for Q supported on the controlled test set. For random coefficients, the same probabilistic convention and coefficient errors must also be controlled.

One sufficient regime is $B_j \leq UV^{j-2}(j-2)!$ with $U, V > 0$ and $\chi = 2RVT < 1$. Then $L_{p,T} \leq 2U/(1-\chi)$ and $\eta_p \leq D\chi^p/p$, where $D = (U/V)e^{2UT/(1-\chi)}$. For $0 < \varepsilon < 1$, put $\lambda_\chi = \log(1/\chi)$. The smallest certified integer order is

$$p_\varepsilon = \max \left\{ 2, \left\lceil \frac{W(\lambda_\chi D/\varepsilon)}{\lambda_\chi} \right\rceil \right\}, \quad M_{\text{NTH}} = \sum_{j=1}^{p_\varepsilon-1} m^j = \frac{m^{p_\varepsilon} - m}{m - 1} \quad (m \geq 2). \quad (45)$$

Here $W(z)e^{W(z)} = z$: the error inequality is equivalent to $(\lambda_\chi p)e^{\lambda_\chi p} \geq \lambda_\chi D/\varepsilon$. The top tensor has m^{p_ε} fixed entries, and direct work is $O(m^{p_\varepsilon})$ per evaluation. Depth is already combined into these kernels. For fixed $m \geq 2$ and χ , the moving count is $\asymp [D/(\varepsilon \log(D/\varepsilon))]^{\log m/\lambda_\chi}$. Tanh analyticity alone does not supply these uniform kernel bounds; $\chi \geq 1$ makes this certificate inconclusive. Population use requires U, V, R independent of the approximation resources. Finite-width initial coefficients also require a width-error allowance; with population coefficients, their construction and approximation are fixed-resource costs.

Scope of the published estimate. After exchanging their width/sample symbols, Huang and Yau (2020, Theorem 2.6) give the training RMS bound

$$\begin{aligned} \frac{\|f(t) - \tilde{f}(t)\|_2}{\sqrt{m}} &\lesssim \frac{(1+t)t^{p-1}}{n^{p/2}} \min\{t, m/\lambda\}, \\ t &\leq \min \left\{ \frac{c\sqrt{\lambda n/m}}{(\log n)^C}, \frac{n^{p/[2(p+1)]}}{(\log n)^{C'}} \right\}. \end{aligned} \quad (46)$$

Time uses the source loss convention; a factor-two time rescaling changes constants, not these powers. This uses their high-probability initialization setting and parameterization, even p , bounded

activation derivatives through $2p + 1$, nondegenerate bounded inputs with uniformly independent subsets through size $2p + 1$, and an initial kernel eigenvalue exceeding λ . Three vectors in two dimensions violate the subset assumption. Their small higher-kernel estimates do not establish (44)'s population hypotheses for our feature-learning flow; their training RMS estimate is not a whole-test-function estimate.

Whole-function evaluation and time analyticity. Set $u_a = -2\tilde{r}_a/m$, $I_\emptyset = 1$, and $\dot{I}_{a_1 \dots a_k} = u_{a_1} I_{a_2 \dots a_k}$ with zero initial values for nonempty words. Repeated integration of the triangular system gives

$$\tilde{f}_p(t, x) = f_0(x) + \sum_{k=1}^{p-1} \sum_{a_1, \dots, a_k} K_0^{(k+1)}(x, x_{a_1}, \dots, x_{a_k}) I_{a_1 \dots a_k}(t). \quad (47)$$

This uses the same number of moving coefficients as the direct hierarchy; one query needs that many contractions *plus* the cost of the fixed mixed-kernel evaluators. When $m = 1$, $I_k = s^k/k!$ and $\dot{s} = u$, so one moving scalar suffices and order dependence remains in fixed coefficients and evaluation. These counts are therefore representation costs, not lower bounds.

Truncation is not a time-Taylor polynomial. For the scalar model $f(\theta) = \theta^2$, loss f^2 , target zero and unit mobility, $f_0 > 0$ gives $\dot{\theta} = -4\theta^3$, $f(t) = f_0/(1 + 8f_0 t)$ and $K^{(j)} = 4^{j-1}f$. Its time-Taylor radius is $1/(8f_0)$, but (47) becomes $\tilde{f}_p = f_0 \sum_{k=0}^{p-1} (4s)^k/k!$, $\dot{s} = -2\tilde{f}_p$. The exact scalar field replaces the sum by e^{4s} , with solution $s = -\frac{1}{4} \log(1 + 8f_0 t)$. On a compact neighborhood of this path for any finite T , the polynomial fields and their derivatives converge uniformly. Subtracting the ODEs gives a vanishing forcing and a uniform Lipschitz coefficient; Gronwall and first exit give convergence on $[0, T]$. Thus a finite time-Taylor radius is no universal barrier for NTH.

E.3 A specified streamed DMFT solver

The sampled self-consistency algorithm of Bordelon and Pehlevan (2022, Appendix B, Algorithm 1) stores layerwise covariance and response kernels indexed by input and time pairs. Each sweep updates predictions, factors Gaussian covariances, samples paths, solves their causal equations and response Jacobians, and averages products and derivatives. Matching its feature-learning scale to ours requires $\gamma = \sqrt{n}$, internal coordinates $U^{(\ell)} = \sqrt{n}W^{(\ell)}$, and the same readout law and loss normalization. A fixed-program Tensor Programs limit (Yang and Hu, 2021, Algorithm 1, Theorem 7.4) does not itself prescribe this numerical solver or a uniform numerical error estimate.

For K time steps, S paths and I sweeps, a dense covariance factor costs $O((mK)^3)$ per layer. Dense triangular solves for a full samplewise response Jacobian cost at most the same per path. Streaming one path and Jacobian while accumulating kernel averages therefore permits

$$M_D = O(Hm^2K^2), \quad C_{D,\text{total}} = O(IH(S+1)m^3K^3), \quad C_{D,\text{step}} = O(IH(S+1)m^3K^2). \quad (48)$$

The last cost is amortized; memory includes kernel and Jacobian workspace. These are implementable upper bounds, not framework-wide lower bounds. Passive test prediction requires mixed kernels and test paths as well.

To justify an error model, suppose the discrete self-consistency map $\mathcal{F}$ is a κ -contraction, $\kappa < 1$, on a common domain, in a norm controlling test outputs, has fixed point z_* , and the sampled map $\hat{\mathcal{F}}$ differs uniformly by at most ζ . For a computed point $\hat{z}$ with residual $\|\hat{z} - \hat{\mathcal{F}}(\hat{z})\| \leq \eta_0$, the triangle inequality gives $\|\hat{z} - z_*\| \leq \eta_0 + \zeta + \kappa \|\hat{z} - z_*\|$. A Lipschitz output map, uniform RMS sampling control and a stable order- r time approximation would thus yield

$$\mathcal{E}_Q \leq A_D S^{-1/2} + B_D (T/K)^r + \eta, \quad \eta \leq C\eta_0/(1 - \kappa). \quad (49)$$

The constants include stability amplification. The cited papers do not establish these hypotheses uniformly as the grid grows for this model. Setting $S = n$ matches a sampling exponent, not the covariance of finite-network fluctuations (Bordelon and Pehlevan, 2023).

Since streaming makes leading memory independent of S , its continuous memory infimum has $K_{\text{inf}} = T(B_D/\varepsilon)^{1/r}$ and $M_{\text{inf}} \asymp Hm^2T^2B_D^{2/r}\varepsilon^{-2/r}$. A finite feasible allocation reserves $\beta\varepsilon$, $0 < \beta < 1$, for sampling and iteration: take $K = \lceil T[B_D/((1-\beta)\varepsilon)]^{1/r} \rceil$, $S \geq (2A_D/(\beta\varepsilon))^2$, and $\eta \leq \beta\varepsilon/2$. Approaching the infimum increases work. Optimizing numerical order requires additional regularity, constants and stage costs; there is no intrinsic DMFT accuracy exponent here.

A distinct finite-width baseline stores every Euler update exactly:

$$W_k^{(\ell)} = W_0^{(\ell)} - \frac{2\Delta}{nm} \sum_{j < k, a} r_{a,j} \delta_{a,j}^{(\ell)} (h_{a,j}^{(\ell-1)})^\top.$$

Its moving memory is $O(HmnK)$, fixed memory $O(Hn^2)$ and late-step work $O(Hmn^2 + Hm^2nK)$. A fixed-stage order- r method has the same storage order; with error $A/\sqrt{n} + D_h(T/K)^r$, optimizing as above gives $M_* \asymp HmA^2TD_h^{1/r}\varepsilon^{-(2+1/r)}$. This stores finite-network updates and is not Tensor Programs sampling.

E.4 A sharper estimate for the original joint clock

Keep exactly $\dot{\tau} = \rho + \|\dot{\Psi}\|_2$, with its matching prefix. Define

$$M_T = 1 + \int_0^T \rho \, dt, \quad V_T = \frac{1}{\sqrt{nm}} \int_0^T \|\dot{\Psi}\|_2 \, dt, \quad \tau_T = M_T + \sqrt{nm}V_T.$$

The insertion measure has mass M_T , while the normalized history $F(\xi) = \Psi(\xi)/\sqrt{nm}$ is $1/\sqrt{nm}$ -Lipschitz.

We need a dimension-independent polynomial approximation fact. If a Hilbert-valued F is L -Lipschitz on $[0, \tau]$, then some polynomial of degree below P has uniform error at most $C_J L\tau/P$. To prove it, extend $G(v) = F(\tau(1 + \cos v)/2)$ evenly and periodically and convolve with the normalized positive trigonometric polynomial $J_N(v) = Z_N^{-1}[\sin(Nv/2)/\sin(v/2)]^4$. On $[-\pi, \pi]$ the ratio is bounded by $\min(N, \pi/|v|)$ and is at least cN near zero. Splitting its integrals at $1/N$ gives $Z_N \asymp N^3$ and $\int |v|J_N(v) \, dv \leq C/N$. The Hilbert norm triangle inequality bounds convolution error by $CL\tau/N$. The convolution is even, hence a polynomial in $\cos v$ of degree at most $2N - 2$. Take $N = \lfloor (P+1)/2 \rfloor$ for $P \geq 2$; a constant polynomial handles $P = 1$.

The weighted projection is no worse than this uniform approximation in $L^2(\mu_T)$. Using the projection energies and product defect of (20) therefore gives

$$\int_0^T \sum_{\ell=2}^H \|E_\ell\|_F \, dt \leq \frac{1}{nm} \sum_{\ell,a} (D_{h,\ell,a} + D_{b,\ell,a}) \leq \frac{C_J^2 M_T}{P^2} \left(\frac{M_T}{\sqrt{nm}} + V_T \right)^2. \quad (50)$$

The first inequality is Cauchy–Schwarz in time followed by $2\sqrt{ab} \leq a + b$; the second uses the actual insertion mass rather than Lebesgue mass τ_T . No clock or algorithm has changed. Width-uniform history mass, normalized variation, feedback stability and continuation thresholds would turn this into a uniform prediction estimate. Establishing those controls, and the matched width rate, remains the population gap.